

Modelling and Control of Heat Distribution in a Powder Bed Fusion 3D Printer

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Master of Science Thesis in Electrical Engineering

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Abstract

This thesis report describes how to improve the control of the temperature in a Powder Bed Fusion 3D printer. This is accomplished by first creating a model of the thermal system. To create a good model, both black-box and grey-box models of the system are estimated and compared. Based on the best model, different control designs are examined and the results are compared to find the control design yielding the best results.

The system being modelled is a multiple input multiple output system with a complex internal structure. The modelling can be divided into several steps. Firstly, data has to be acquired from the system. Secondly, the data is analysed and processed. Thirdly, models are estimated based on the collected data. Different model structures such as state-space, ARX, ARMAX, Output Error, Box Jenkins and grey-box models are examined and compared to each other. Finally, the different derived models are validated and it turns out the ARMAX model yields the best prediction capabilities. However, when the controllers were tested on the actual system the controllers that are based on the grey-box model yield the best results.

The different control designs examined in this work are diagonal PI controllers, decoupled PI controllers, feed forward controllers, IMC controllers and state feedback controllers. The controllers are all based on the derived models.

The controllers are implemented into a code structure capable of communicating with the printers. Here, tests of the performance for the different controllers on the actual system are executed. The results show that a non-linear system can be controlled using linear controllers. However, introducing some fuzzy control elements such as limiting the controllers to only be used within small temperature intervals and using a fixed input outside this interval yield better results. From these results, the best linear controller is a diagonal PI controller tuned from a grey-box model with as many states as there are controllable areas of the powder bed. The improvement is only marginal compared to the original PI controller, reinforcing the conclusion that some non-linear strategies are needed in the controller in order to achieve significant improvements.

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Acronyms

ARMAX Autoregressive-moving-average with exogenous terms. *Glossary:* AR-MAX

ARX Autoregressive with exogenous terms. *Glossary:* ARX

BJ Box Jenkins. *Glossary:* BJ

IMC Internal Model Control. *Glossary:* IMC

LQG Linear Quadratic Gaussian. *Glossary:* LQG

MIMO Multiple Input Multiple Output. *Glossary:* MIMO

MPC Model Predictive Control. *Glossary:* MPC

MSE Mean Square Error. *Glossary:* MSE

OE Output Error. 34, *Glossary:* OE

PA Polyamid. *Glossary:* PA

PBF Powder Bed Fusion. *Glossary:* PBF

PEM Prediction-Error Method. *Glossary:* PEM

PRBS Pseudo Random Binary Signal. *Glossary:* PRBS

RGA Relative Gain Array. *Glossary:* RGA

SITB System Identification Toolbox. *Glossary:* SITB

SLS Selective Laser Sintering. *Glossary:* SLS

SPA Spectral Analysis. *Glossary:* SPA

1

Introduction

The last couple of years has seen an explosive increase in usage and the demand of 3D printers. One company that is trying to meet these demands is Wematter. Wematter develops Powder Bed Fusion (PBF) based 3D printers. PBF is an additive manufacturing process that fuses selective parts of a powder base layer by layer to build an object. In order to create products of better quality, more concise control of the different temperatures in the printer is needed.

This chapter will provide a description of the problem and a brief hardware description. Furthermore, the objective of the thesis work will be clearly outlined. Different limitations that are imposed will also be explained. Thereafter, the method and approach that was used will be justified. Finally, the outline of the thesis report will be presented.

1.1 Hardware and Problem Background

The typical structure of a PBF printer can be seen in Figure 1.1 [27]. The material used for printing is during this work a Polyamid (PA) powder that melts and fuses at known temperatures.

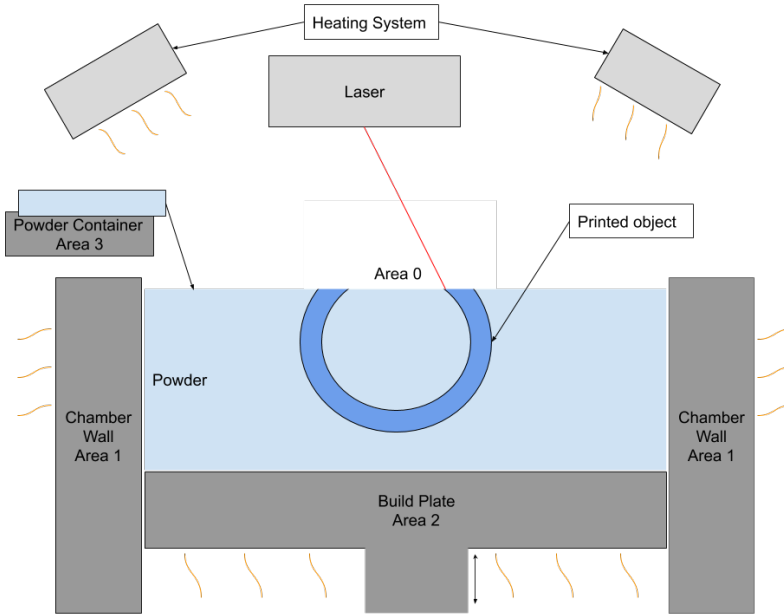


Figure 1.1: Typical PBF 3D Printer structure.

There are four areas of heating elements and sensors in the 3D printer used in this thesis project. Some of the temperature sensors are known as thermocouples, which produce a temperature-dependent voltage. The different areas all affect each other and to fully control the temperature of the system all areas must be controlled.

Area 0: Optical heating system - during-print heating The powder needs to be at a stable temperature below the melting point. To achieve this, the powder is heated from above through an optical heating system with multiple channels that can be used to control the power to heaters affecting different parts of the powder base. **The powder base is divided into N controllable areas.** The powder temperature is measured using multiple IR-based pyrometers which measure at different parts of the powder bed.

Area 1 and 2: Conductive heating system - post-print heating When a layer has been printed, the build plate is lowered to allow for additional powder to be placed on top of the previous layer. In order to keep this lowered powder from cooling too quickly, a number of conductive heating elements are placed on the chamber walls (see Figure 1.1), and on the build plate in order to heat the volume. The heating elements on the wall are controlled using one channel and the heating elements in the build plate are controlled by another channel. The temperatures in these areas are measured using two thermocouples.

Area 3: Conductive heating system - pre-print heating Before the powder is de-

livered into the printing volume, there is an option to preheat it in its container. The temperature in this area is measured using a thermocouple.

Originally area 1, 2 and 3 all have identical PI controllers that control the temperatures independently from each other. Similarly, the original solution for area 0 has N identical PI controllers, one for each of the N controllable powder areas.

1.2 Problem Formulation

To improve the temperature control in the 3D printer, a model of the system is needed. The model should be of Multiple Input Multiple Output (MIMO) character in the sense that it takes the correlations and biases between the heat sources and the IR sensors into account. With this in place, a suitable control law can be created and implemented. Hence, better temperature control for the printer can be achieved.

It is essential that the derived controller is functional under realistic operational conditions. Therefore, realistically numerous disturbances will continuously be affecting the system controlled by the derived temperature controller. Some of the disturbances are temperature fluctuations, temperature leakage, new powder being applied and a laser selectively heating up different areas.

In order for the thesis work to be considered successful by the company, the resulting prints need to have less discoloration than the current design. To achieve this, the controller must keep the printing medium within a relatively small temperature window and maintain a smooth temperature gradient throughout the medium. The controller should be able to handle a change of printing powder and still yield reasonable results. A bonus goal is that the derived controller makes the entire printing process faster than the solution developed by Wematter.

1.3 Objectives

The desired and expected result from this thesis work is a parametric model of the heat distribution that enables the implementation of a controller that yields better results than the current solution. Better is in the sense of less temperature fluctuations.

In conclusion, the questions this thesis work aims to answer are:

- Can the system be described by a linear parametric model?
- Can the temperature of the 3D printer be controlled using a controller based on a MIMO model?
- Does the use of the aforementioned controller result in better temperature control than the original solution does under realistic conditions as

described in Section 1.2?

1.4 Limitations

This thesis work only examines the 3D printer *Gravity* developed by Wematter. Hence, all results are only applicable to this printer and no general conclusions are made.

Due to the company's desire to keep most of their design a secret, this report will not contain specifics regarding the amount of sensors in the machine or go into details of mechanical parts that are patented.

1.5 Method

The overall method and scientific approach to solve the thesis problem is based on dividing the work into modelling, control design and finally testing. By testing different model structures and modelling approaches based on the collected data, a good fundamental understanding of the system can be achieved. Furthermore, based on the models, different control designs are developed and their functionality is tested through simulations. Finally, the different derived controllers are tested on the actual system and properly validated.

1.6 Thesis Outline

The report begins with a chapter briefly presenting the theoretical background needed for the work, see Chapter 2. Thereafter, Chapter 3 presents the modelling that is done in the thesis work. The chapter contains both black-box models, grey-box models and validations of the different models. Furthermore, Chapter 4 describes the different control designs that have been developed to better control the temperature. In Chapter 5 the different models and control designs are compared and validated on the real system. Based on the validation the optimal model and controller are chosen. After that, Chapter 6 contains the authors reflections surrounding the work and Chapter 7 lists the conclusions drawn during the work. Possible future work are finally presented in Chapter 8.

2

Modelling and Control Design Theory

This chapter describes the underlying theory used to create the different models. The theoretical foundation of the thesis can be divided into two parts: modelling and control design.

2.1 Modelling

Modelling is the scientific method to describe a system mathematically. Two different approaches to describe a system will be explained. First black-box modelling and second grey-box modelling.

2.1.1 Black-box Modelling

For systems being modelled with black-box models no physical model of the systems exist beforehand. The reason can either be that the system is assumed to have a very complex internal structure or that no time or knowledge exist to create the physical model. Hence, for black-box models, no information other than the input and output of the system exists to estimate a model from. System identification using different black-box models has been widely studied the last decades. Therefore, different parametric structures have been developed that often manage to capture the essential dynamics of complex systems [16].

The commonly used linear parametric models and their characteristic features will now be presented. Autoregressive with exogenous terms (ARX) is the simplest model, and is often used in accordance with TSTF, try simple things first. The model is described by $A(q)y(t) = B(q)u(t) + e(t)$.

The ARX model structure can often be improved by adding a more complex noise

model, resulting in an Autoregressive-moving-average with exogenous terms (ARMAX) model. It is similar to the ARX model but considers the noise through the moving average. The model is described by $A(q)y(t) = B(q)u(t) + C(q)e(t)$.

Moreover, the ARX model can be modified into a model structure that ignores modelling any of the noise of the system. This is the model known as the Output Error (OE) model. The model solely describes the relation between the output and input signal and can be written $y(t) = \frac{B(q)}{F(q)}u(t) + e(t)$.

Finally, the OE model structure with further modelling of the noise can be developed into a more general structure, known as the Box Jenkins (BJ) model. The model is described $y(t) = \frac{B(q)}{F(q)}u(t) + \frac{C(q)}{D(q)}e(t)$.

All of these models are based on the same family of model structures [15] which can be described by the generalised model equation

$$A(q)y(t) = \frac{B(q)}{F(q)}u(t) + \frac{C(q)}{D(q)}e(t) \quad (2.1)$$

where $y(t)$ denotes the measured output, $u(t)$ the control input and $e(t)$ the system noise (assumed Gaussian in most cases). $A(q)$, $B(q)$, $C(q)$, $D(q)$ and $F(q)$ are polynomials.

The models can for example be estimated through a Prediction-Error Method (PEM) where the prediction error is defined as

$$\epsilon(t, \theta) = y(t) - \hat{y}(t|\theta) \quad (2.2)$$

. The predictor $\hat{y}(t, \theta)$ can be derived from (2.1) and yields

$$\hat{y}(t, \theta) = \frac{D(q)B(q)}{C(q)F(q)}u(t) + \left[1 - \frac{D(q)A(q)}{C(q)}\right]y(t) \quad (2.3)$$

The estimated parameter value can be obtained by minimising $V_N(\theta) = \frac{1}{N} \sum_{t=1}^N \epsilon^2(t, \theta)$ i.e., $\hat{\theta}_N = \operatorname{argmin}_{\theta} V_N(\theta)$.

For a MIMO system $A(q)$, $B(q)$, $C(q)$, $D(q)$ and $F(q)$ in (2.1) become matrix polynomials and the presented theory can be extended to hold also for this case [15].

2.1.2 Grey-box Modelling

Grey-box modelling requires a priori knowledge about the system or process examined [26]. The amount of knowledge can vary, but the unknown dynamics of the system are estimated using parametric models as in the black-box approach above. One advantage of using a grey-box model compared to using a black-box model is the possibility to exclude non-linear phenomena from the estimated part of the model. Naturally, this requires the non-linear phenomena to be described in the physical part of the model.

To create a general physical model and its corresponding state-space model, a method that utilises analogies between different physical domains can be used. Different physical systems, such as thermal, electrical, mechanical, etc., can all be modelled by examining different efforts and flows. The resulting description is called bond graphs. Chapter 6 in [16] provides a thorough explanation of the subject.

Table 2.1 illustrates physical attributes that are correlated between the electrical and thermal domains.

Table 2.1: Analogies between domains

	Electrical	Thermal
Flow	current	heat flow
Effort	voltage	temperature
Power	power	power $\times$ temperature
Capacitive Element	capacitor	heat storage
Resistive Element	resistance	thermal resistance

The basic electrical and thermal equations needed for the bond graphs are:

$$C \frac{dU(t)}{dt} = I(t) \quad (2.4a)$$

$$mc_p \frac{dT(t)}{dt} = \dot{Q}(t) \quad (2.4b)$$

$$I(t) = \frac{1}{R} U(t) \quad (2.5a)$$

$$\dot{Q}(t) = \frac{kA}{l} \Delta T(t) \quad (2.5b)$$

where C is the a capacitance, U is the voltage, I is the current and R the resistance. For the thermal equations m is the mass, c_p the specific heat capacity, T the temperature, Q the heat flow, k a proportional constant, l the distance between the objects and A the heat transfer area [18].

Equation (2.4a) and (2.4b) are correlated, as well as (2.5a) and (2.5b).

Bond graphs are constructed by creating a network where the efforts and flows of the system are described. Furthermore, different nodes in the physical model are modelled either as a series connection or a parallel connection. From the bond graph and domain specific physical equations, a state-space model can be deduced.

2.2 Control Design

The system as a whole has four types of control signals as described in Section 1.1. They control different types of heaters in different environments which result in vastly different dynamics. In theory, they could be coordinated and used together to reach the goal temperature on the powder bed in for example a cascade controller. However, since each of the four areas have their own reference temperature, this becomes impractical and unnecessarily complicated and would require a more advanced controller such as a Split Range Controller or Model Predictive Control (MPC). Instead, each controller controls the temperature of their respective printer area independently from each other.

The focus of this thesis work lies on the temperature of the powder bed and the control of the optical heating system. Here, different control strategies are used in different iterations. The first iteration utilised a diagonal PI controller, where the controller $F(s)$ is on the form

$$F(s) = \begin{pmatrix} F_1(s) & 0 & \cdots & 0 \\ 0 & F_2(s) & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & F_n(s) \end{pmatrix} \quad (2.6)$$

. The standard feedback controller can be seen in Figure 2.1.

In Section 2.2.1 it was discussed how different models are discretized. The different block diagrams being described in this work usually illustrates continuous-time connections. However, when they are implemented they have of course been discretized.

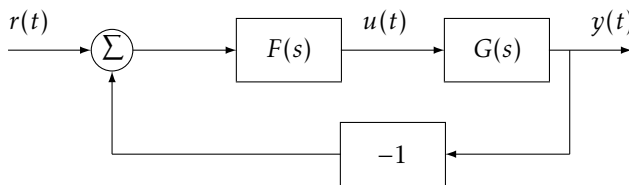


Figure 2.1: A feedback controller.

2.2.1 Discretization of Continuous-Time Systems

Discretization of a continuous-time system is necessary for digital implementation. The signals are transformed to the discrete time form, $y(t) \rightarrow y(kT_s)$, $r(t) \rightarrow r(kT_s)$ and $u(t) \rightarrow u(kT_s)$, where $k = 0, 1, \dots$ and T_s is the chosen sampling time.

The continuous-time operation of derivation can be approximated using various methods, for example the Euler methods or the Tustin method [9]. To simplify notation, the delay operator q^{-1} , which is defined as

$$q^{-1}u(kT_s) = u(kT_s - T_s) \quad (2.7)$$

is introduced. The Euler backward approximation is now described as

$$p = \frac{1}{T_s}(1 - q^{-1}) \quad (2.8)$$

, and the Tustin approximation as

$$p = \frac{2}{T_s} \frac{(1 - q^{-1})}{(1 + q^{-1})} \quad (2.9)$$

where p is the continuous-time differential operator defined as $pu(t) = \dot{u}(t)$. Using the methods described in this section, a continuous-time transfer function can be transformed to a discrete-time transfer function.

2.2.2 Tuning of PI or PID Controller

Usually, one can achieve surprisingly good results using only a well fitted PID controller. The first step when aiming to improve an existing diagonal PID controller is to tune it for the system being controlled. For this, there exists many approaches that stem from a three parameter model,

$$G(s) = \frac{K_p}{1 + sT} e^{-sL} \quad (2.10)$$

, estimated from a step response of the system [9]. The parameters are K_p which is the static gain of the system, T which is the step time and L which is the system lag. Approaches that do not stem from a three parameter model exist as well, but is not used in this thesis work due to the three parameter model yielding sufficiently accurate approximations.

2.2.3 Controller Decoupling

By doing a Relative Gain Array (RGA) analysis on the acquired model of the system, the relations between input and output signals become more tangible. The RGA is defined by

$$RGA(G) = G \circ G^{-T} \quad (2.11)$$

where G denotes the model of the system and the operator "o" implies element-wise multiplication. A good pairing yields elements in the RGA matrix close to 1, which also indicate that the input signals mostly only affect one output signal each. In cases where the elements in the RGA matrix are not close to 1, one might consider decoupled control to simplify the design of the controller and keep the diagonal design.

With this, the controller assumes the form

$$F(s) = W_1 F^{diag}(s) W_2 \quad (2.12)$$

where W_1 and W_2 are commonly chosen as

$$\begin{cases} W_1 = G(0)^{-1} \\ \text{or} \\ W_1 = G(w_c i)^{-1} \end{cases} \quad \text{and} \quad W_2 = I \quad (2.13)$$

. Here, w_c denotes the cut-off frequency. This controller results in a circuit gain that is approximately diagonal.

In this thesis work however, W_1 has been chosen according to

$$W_1 = (CB)^{-1} \quad (2.14)$$

with the motivation that this should compensate for any non diagonal dependencies from input to output. Here, C and B represent matrices describing the dynamics of G when expressed in state space form. This modification yields a faster controller compared to the first alternative in (2.13) due to the overall larger gain in the resulting matrix.

2.2.4 Feed Forward from Reference Signal

Using a sufficiently accurate model of the system, it might be advantageous to use a feed forward controller as illustrated in Figure 2.2.

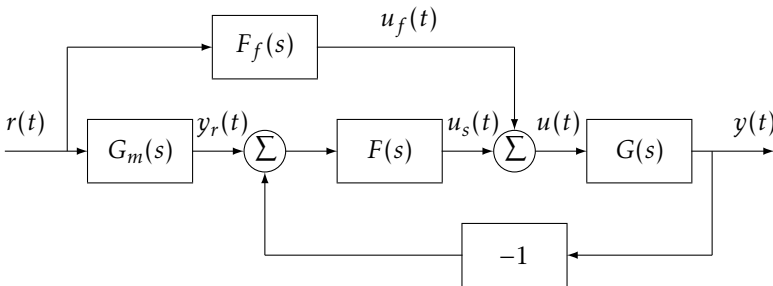


Figure 2.2: A feed forward controller.

This controller can for example yield step responses that are faster than that of a PID controller, whilst maintaining good stability margins [9].

The feedback controller $F(s)$ remains the same as in the previous step in Chapter 2.2.2 or Chapter 2.2.3. $G(s)$ still denotes the system which we cannot change. $G_m(s)$ and $F_f(s)$ however, we choose in order to yield desired results. Here, there are two common but distinctly different methods to choose G_m while F_f is designed the same in both methods, according to

$$F_f(s) = \frac{G_m(s)}{G(s)} \quad (2.15)$$

. These are the Ideal feed forward and the Neutral feed forward methods [9].

With Ideal feed forward control, we assume we have a model close to the actual system. We want the closed loop system to be equal to G_m , leaving the feedback controller with as little as possible to do. Therefore, we design G_m with the properties we want the closed system to have, for example to have a step response with a certain time constant T as shown in

$$G_m(s) = \frac{1}{1 + sT} \quad (2.16)$$

. Note however that G_m should have the same relative degree as G to ensure that F_f is causal and proper.

Neutral feed forward is desired where the open system yields good enough step responses. Here, G_m is designed according to

$$G_m(s) = \frac{G(s)}{G(0)} \quad (2.17)$$

.

2.2.5 Internal Model Control

Another method is to use the model of the system directly in the controller. This is the Internal Model Control (IMC) -method and it is illustrated in Figure 2.3. The idea is to only use the difference between the real system, $G_0(s)$, and the model, $G(s)$, as feedback in the system. With an entirely accurate model, i.e. $G(s) = G_0(s)$, the resulting closed system becomes

$$Y(s) = G(s)Q(s)\tilde{F}_r(s)R(s) \quad (2.18)$$

The ideal choice of $Q(s)$ would be $Q(s) = 1/G(s)$, but, this is practically impossible and $Q(s)$ will have to be modified [9]. In [11], the authors give some examples on how this can be done depending on the properties of $G(s)$. $\tilde{F}_r(s)$ is chosen such

that the reference signal take on the desired properties.

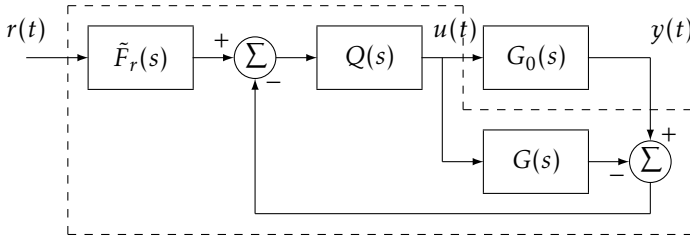


Figure 2.3: A IMC controller.

2.2.6 State Feedback Control

A controller based on feedback from estimated states use the model to predict how the states of the system change with time and control input. The general structure can be seen in Figure 2.4 where L is a feedback matrix. The matrix L and the observer matrices can be chosen in several different ways, for example by pole placement or by utilising Linear Quadratic Gaussian (LQG) control. L_0 is chosen to compensate for the static gain of the system [11].

LQG control finds the controller through minimisation of a quadratic problem and assumed Gaussian noise. By minimising

$$\min(\|e\|_{Q_1}^2 + \|u\|_{Q_2}^2) = \min \int e^T(t)Q_1e(t) + u^T(t)Q_2u(t) \quad (2.19)$$

, the feedback matrix L can be derived [11].

Q_1 and Q_2 are design parameters that are chosen by the user. Similarly the observer is chosen by minimisation of a quadratic problem and tuning of design parameters [11, 13].

A state feedback controller can be extended with an external integrating controller. This might be needed if there are constant disturbances or if the model is not accurate enough [10].

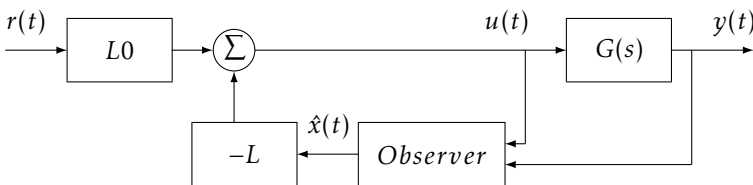


Figure 2.4: A state feedback controller.

3

System Identification

Creating a mathematical description of a dynamic system is the essence of system identification. The temperature fluctuations in a 3D printer are numerous and they are hard to describe physically. Therefore, two different modelling approaches are examined and compared in this chapter, black-box modelling and grey-box modelling. Furthermore, the data acquisition and the validation of the different models will be presented.

3.1 Data Acquisition

System identification is fully dependent on the data that the models are based on. Therefore, a crucial step in model estimating is to acquire informative data.

3.1.1 Choice of Input Signal

The choice of input signal is essential for the data acquisition. Since an open loop experiment was used, the input was considered accordingly, and a Pseudo Random Binary Signal (PRBS) was chosen. A good input signal has to excite all the modes of the system, by containing a sufficient amount of relevant frequencies. One important signal property to monitor is the Crest Factor

$$C_r^2 = \frac{\max_t (u(t) - 0.5)^2}{\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N (u(t) - 0.5)^2} \quad (3.1)$$

where $u(t)$ is the input signal and 0.5 is the median input value. Theoretically we want to minimise the Crest Factor [15]. The lower bound for the crest factor is 1 and can be achieved with a binary signal. A good signal has a small crest factor. However, the spectrum of the input signal must also be considered. L. Ljung defines the problem in [15] as to "achieve a desired input spectrum for a signal

with as small crest factor as possible". By examining in which frequency band a system is excited, the interesting part of the spectrum can be found. From Figure 3.1 the interesting frequency is decided to be around 100 [mHz] since there is a noticeable bump in this region.

The input signal is adjusted so that its input power is focused in the relevant frequency band. This type of signal is chosen since it yields a low crest factor and allow for input frequency manipulation[15]. The amplitude of the input signal is between $0 \leq u(t) \leq 1$, where $u(t) = 0$ yields no power to the optical heaters and $u(t) = 1$ yields full power to the optical heaters. Hence, the result is a crest factor close or equal to 1. The spectrum of the input signal can be seen in Figure 3.2. Note that each input signal uses a unique PRBS signal. They all have the same properties, but are generated independently from each other as to not introduce any correlation in the system input.

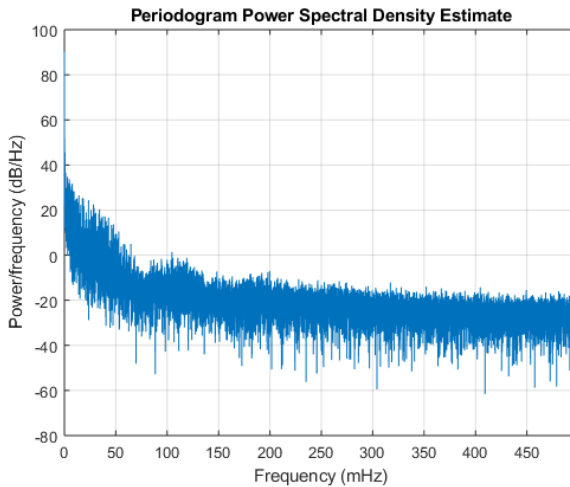


Figure 3.1: Periodogram of the output signal power spectral density expressed in Hertz.

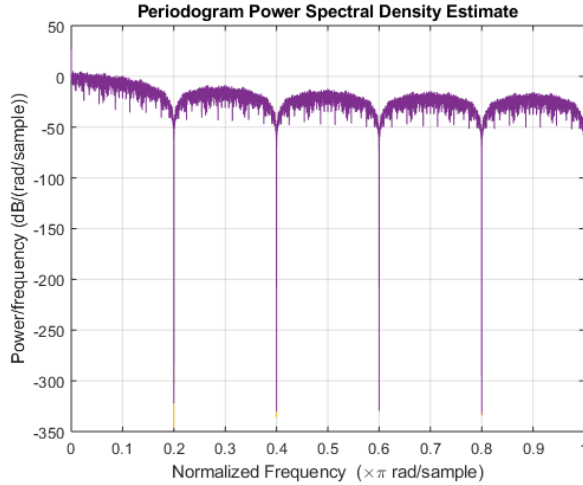


Figure 3.2: Periodogram of the input signal power spectral density expressed in normalised frequency.

From old data of the system, the time constant τ was estimated to be at least $\tau > 10$ [s]. Therefore, a sampling time of $T_s = 1$ [s] was deemed sufficient. Aliasing is avoided by choosing a high sampling frequency, and the sampling is more than quick enough to capture the relevant system dynamics.

3.1.2 Pre-processing of Data

The captured data is processed before it is used to create a model. Initially, the data is cropped so that only the relevant parts of the data are analysed. This removes data from the pre-heating and the cooldown phase, as well as any data gathered in the test phase when the controller is activated due to too high or low temperatures. Furthermore, as the printing progresses, the printer accumulates heat internally, leading to a higher temperature gain for the same input power. Hence, the data is detrended to mitigate any effects this might introduce. The detrending removes both the mean of the signal and any linear trends that might exist in the experimental data.

As a final step, the data is filtered using a high-pass filter to remove any remaining slow trends in the data. Extensive information regarding how to filter signals can be found in both [14] and [19]. This results in the power spectrum in Figure 3.3. The high-pass filter is needed because the detrending is not able to fully mitigate the fact that the system slowly accumulates heat as it is active. The accumulations results in non-linearities in the temperature output. The model should not try to model any slow non-linearities and the high-pass filter is therefore needed to ensure that the created model is only modelling the relevant behaviour of the system.

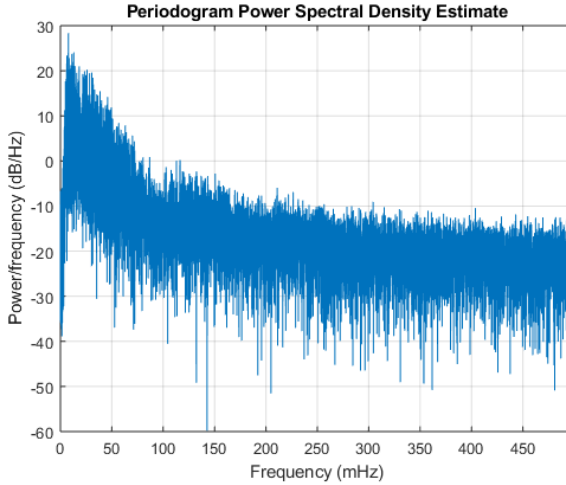


Figure 3.3: Periodogram of high pass filtered output signal power spectral density expressed in Hertz.

3.2 Black-box Modelling

The models examined in this chapter are state-space models, transfer function models and polynomial models. These models are generally estimated by using a PEM.

From the data, different structures described by (2.1), are examined. Through the use of the System Identification Toolbox (SITB) in Matlab, different model structures are estimated [17].

3.2.1 Model Order Selection

An important part of the model estimation is to limit the complexity while still modelling all the important dynamics of the system. This is done by choosing the order of the polynomials in the models. A rough estimate or starting point is given by the Matlab function *n4sid*, that evaluates a range of orders of state-space models and picks the best order from a Hankel Singular Value plot[17]. As illustrated in Figure 3.4, it may be appropriate to start with order six.

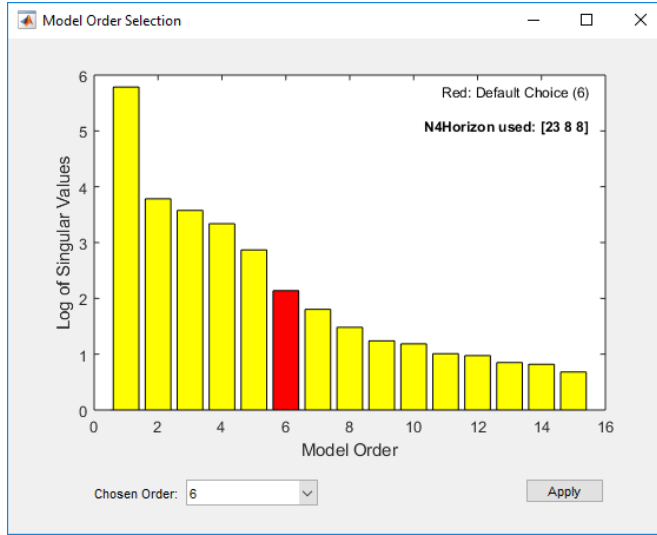


Figure 3.4: Model order selection using the Matlab function *n4sid*.

3.2.2 Estimating Input Delay

A basic step when designing models is to estimate the input delay of the system, i.e. to estimate the time it takes for the input to affect the output. This is measured in lags where one lag is the time between one sample and the next sample.

To get an indication of the magnitude of the input delay in the 3D printer, an impulse response was estimated from the captured data and plotted as Figure 3.5. The light blue area indicates a 97% confidence interval. The illustrated impulse response was estimated using the state-space model of order six calculated in Section 3.2.1. The figure clearly illustrates how the impulse response takes off and leaves the confidence interval almost instantly, indicating that the system does not include any large time delay. With a lag of somewhere between zero and one, the choice is made to create the models assuming a lag of zero. If the system truly has a lag of one, then the models will simply ignore the term of order zero.

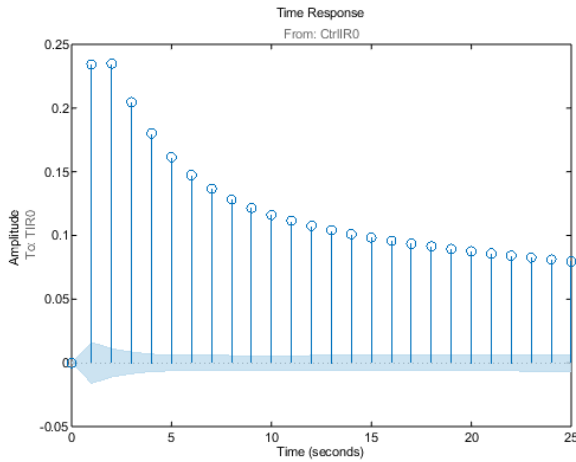


Figure 3.5: Impulse response with confidence interval illustrating the lack of a time lag in the system.

3.2.3 Model Type Selection

By using the recommended model order from above as starting point, a number of model types are estimated by minimising the one-step ahead prediction error between measured and predicted outputs. These models are compared with each other and with validation data for one of the outputs, as seen in Figure 3.6. From this, it seems that most model types can describe the system fairly well. However, due to some unreliability when changing between data sets, the state-space, ARX and ARMAX models are deemed the most interesting and viable for this application and will be examined further in Section 3.5. The model order is 6 for each model presented in Figure 3.6, and the choice of order will be further discussed in Section 3.4.1. Figure 3.6 contains one of the output signals, but in reality the output is multidimensional.

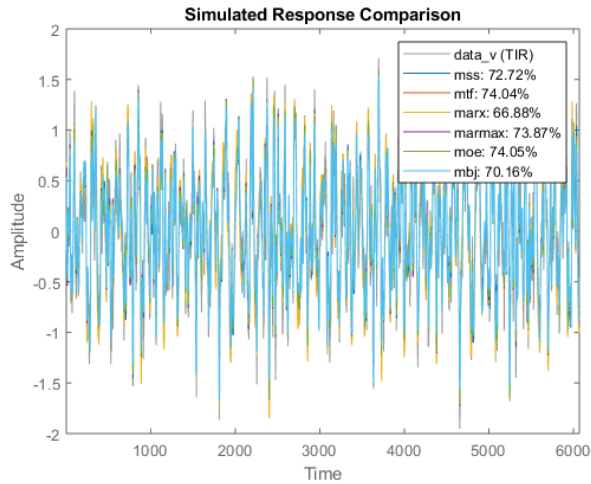


Figure 3.6: Several models compared to validation data for one of the outputs.

The averaged model fits over all the areas for the different models can be seen in Table 3.1 and shows that all models except the BJ and OE models perform well in all areas of the printer.

Table 3.1: Model fit average over all areas for the investigated models.

State-space	Transfer Function	ARX	ARMAX	OE	BJ
73%	59%	66%	74%	32%	-558%

3.3 Grey-box Modelling

Physical models are used for the grey-box modelling. Different physical models have been developed in this project and tested to find the optimal model. In this section only one model will be fully presented. However, in Appendix A other derived models are presented for the avid reader. The inspiration for this modelling approach can be found in [21] and [20].

The model is based on the system illustrated in Figure 3.7. The powder bed being modelled is, as previously mentioned in Section 1.1, divided into N areas. One area, i , is connected to another area, j . The connection between the different areas is assumed, in accordance with observations from the data, to be well isolated and any heat exchange between two areas is therefore assumed negligible. Each modelled area has a corresponding controllable heating element. However, each heating element will have a significant effect also on the other areas. The arrows in Figure 3.7 illustrate how the heating element of one area affects other

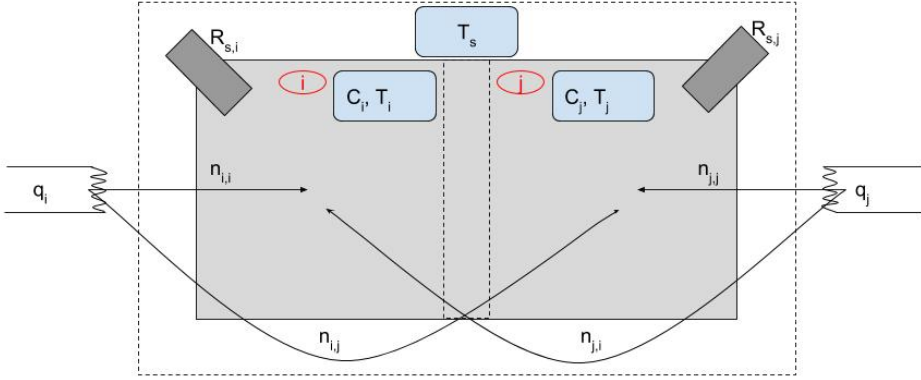


Figure 3.7: Model of thermal system with N locations being modelled.

areas, and the effect is modelled by a factor $n_{i,j}$ corresponding to each arrow. Furthermore, each area has a heat storage modelled as a capacitor and a thermal resistance connected to the surrounding temperature.

The input to the system, modelled as q_i in Figure 3.7, is in reality an electric current. Hence, a transformation between different physical domains is needed in the model. Therefore, the system can conveniently be described by a bond graph. Figure 3.8 illustrates part of the bond graph for the system.

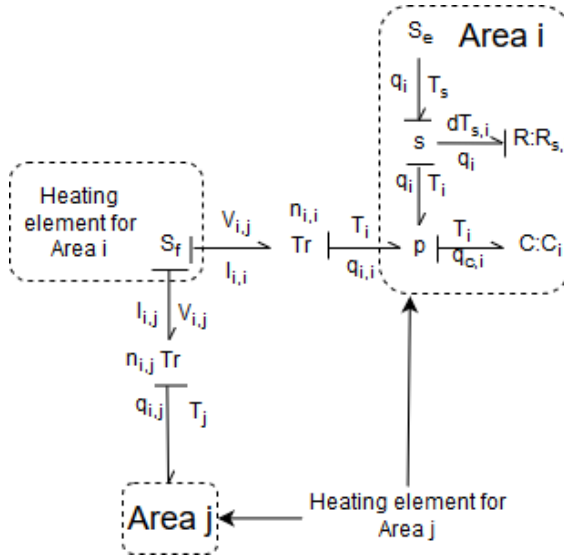


Figure 3.8: Bond graph of thermal model.

From Figure 3.8 and (2.4b), one can derive

$$q_i = \frac{1}{R_{s,i}}(T_s - T_i) \quad (3.2)$$

, which describes how the surrounding temperature affects the different areas. The N heating elements are controlled electrically and the transformations between the domains are done using a transformer. The scaling from electrical current to the heat effect measured by the state is modelled by the $n_{i,j}$ variables. Physically, a transformation represents the transformation from electrical energy to thermal energy and the effect that a specific heating element has on a specific temperature being measured. Furthermore, the heat flow for an area can be modelled by

$$C_i \dot{T}_i = q_i + \sum_j q_{i,j} \quad (3.3)$$

, where, $i \in [1, \dots, N]$ and $j \in [1, \dots, N]$. Note that the heat input is described by $q_{i,j} = n_{i,j} I_i = n_{i,j} u_i$.

Combining (3.2) and (3.3) results in

$$C_i \dot{T}_i = \frac{1}{R_{s,i}}(T_s - T_i) + \sum_j q_{i,j} \quad (3.4)$$

.

Equation (3.4) can be extended into a state-space model. Choosing $x = [T_1, \dots, T_N]^T$, $u = [I_1, \dots, I_N]^T$ and $w = T_s$ result in

$$\dot{x}_i = \frac{-1}{C_i R_{s,i}} x_i + \sum_j \frac{n_{i,j}}{C_i} u_j + \frac{1}{C_i R_{s,i}} w \quad (3.5)$$

. I_i , T_i and T_s can be measured.

When writing the state-space model on standard form,

$$\begin{aligned} \dot{x} &= Ax + Bu + N_s w \\ y &= Cx \end{aligned} \quad (3.6)$$

, the A , B , C and N_s matrices are given by

$$A = \begin{pmatrix} \frac{-1}{C_1 R_{s,1}} & 0 & \cdots & 0 \\ 0 & \frac{-1}{C_2 R_{s,2}} & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{-1}{C_N R_{s,N}} \end{pmatrix} \quad B = \begin{pmatrix} \frac{n_{1,1}}{C_1} & \frac{n_{1,2}}{C_1} & \cdots & \frac{n_{1,N}}{C_1} \\ \frac{n_{2,1}}{C_2} & \frac{n_{2,2}}{C_2} & & \frac{n_{2,N}}{C_2} \\ \vdots & & \ddots & \vdots \\ \frac{n_{N,1}}{C_N} & \frac{n_{N,2}}{C_N} & \cdots & \frac{n_{N,N}}{C_N} \end{pmatrix} \quad (3.7)$$

(3.8)

$$C = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \quad (3.9) \quad N_s = \begin{pmatrix} \frac{1}{C_1 R_{s,1}} & 0 & \cdots & 0 \\ 0 & \frac{1}{C_2 R_{s,2}} & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{C_N R_{s,N}} \end{pmatrix}. \quad (3.10)$$

The A matrix, seen in (3.7), can be seen to be fully diagonal, indicating that no state is affecting any other state, which corresponds to the previous assumption that the states seem to be well isolated from each other. On the other hand, the B matrix, seen in (3.8), can be seen to be non-diagonal. The estimated values of the different $n_{i,j}$ will give a clear picture of how the input of the system is distributed. The N_s matrix describes the effect that the surrounding temperature has on the system.

The derived model is a well structured state-space model of order N . The B matrix in particular will be very informative of how the heat distributes from the heat elements to the different states.

3.3.1 Model Estimation

From the derived structure the different parameters are estimated by minimising the one-step ahead prediction error similarly to how the parameters of the black-box model parameters are estimated. Different start values for the optimisation problem are tested and the ones yielding the optimal results are chosen. The resulting model output compared to validation data for one of the output channels can be seen in Figure 3.9.

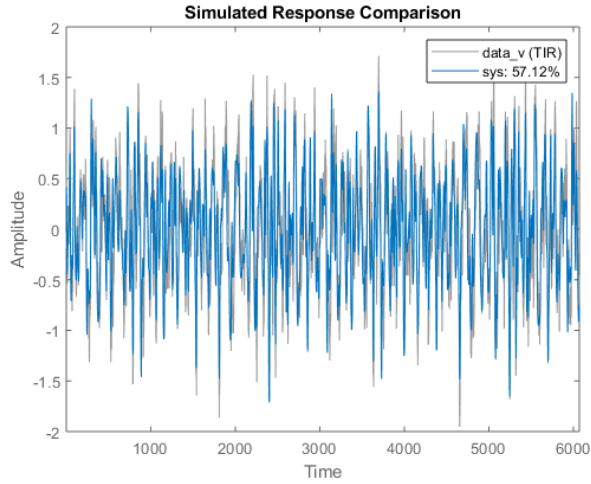


Figure 3.9: Model comparison to validation data for one input/output transfer function for the grey-box model.

The average model fit over all the areas is 56%.

3.4 Model Validation

This section describes the validation done for the state-space, ARX and ARMAX models. It is important that the different models obtained are compared and validated so that the best model is chosen. The validation differs between black-box and grey-box modelling in the sense that the estimated parameters in the grey-box model corresponds to real physical parameters, thus having some physical limits.

3.4.1 Black-box Model Validation

Different aspects are examined when validating a black-box model, e.g. output fit ratio, estimated step response, Bode diagram, zeros and poles placement and residuals.

When looking at the step responses, some model structures and model orders seem to produce models where the diagonal elements are not dominant for some input-output combinations. One example of this is shown in Figure 3.10 where an ARMAX model of order 6 is used. Here it is clear that the first input signal has the largest effect on both output signals. This not only makes the system harder to control, but it does not quite reflect observations made on the physical system during tests. Hence, models displaying this kind of behaviour are dismissed.

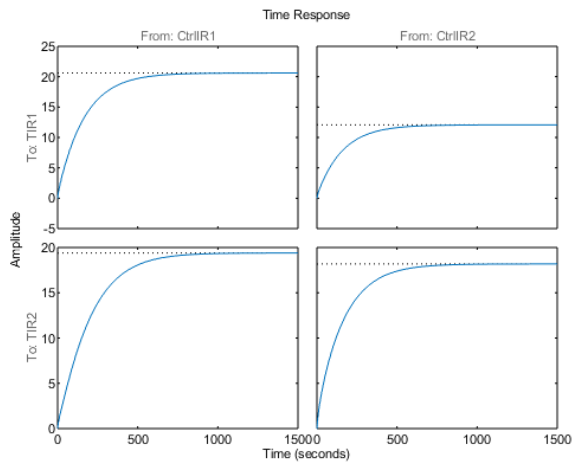


Figure 3.10: Estimated step response of ARMAX model of order 6 on two input-output combinations.

A more desirable step response is illustrated in Figure 3.11 where the diagonal elements are, at least somewhat, dominant.

Table 3.2 shows the model fit of the models with reasonable step responses. It is

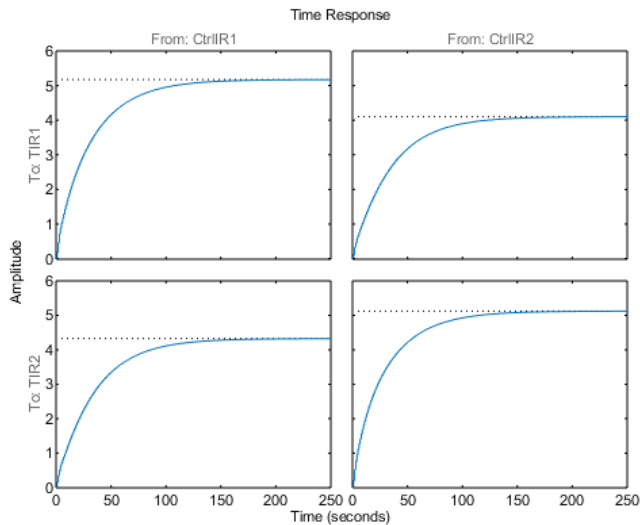


Figure 3.11: Estimated step response of ARX model of order 6 on two input-output combinations.

Table 3.2: Average fit percentage of models with acceptable step responses.

Model Type	Model Order	Fit percentage
state-space	4	57
state-space	3	56
ARX	11	71
ARX	10	70
ARX	9	69
ARX	8	69
ARX	7	67
ARX	6	66
ARX	5	64
ARX	4	61
ARX	3	58
ARMAX	2	57

quite clear that the ARX model structure shows most robustness with regard to realistic step responses.

The Bode diagram for one input/output relation for some of the above mentioned models is presented in Figure 3.12. The different models are compared to a Spectral Analysis (SPA) frequency response, specifically a Blackman-Tukey spectral analysis. The comparison is useful since the models and the spectral analysis are derived from relatively different presumptions [15]. From Figure 3.12 it can be observed that all models capture the essentials in the magnitude plot. The most impactful differences occur at frequencies close to the Nyquist frequency where the high order ARX models of order 6 and 10 follow the frequency dip better than the other models.

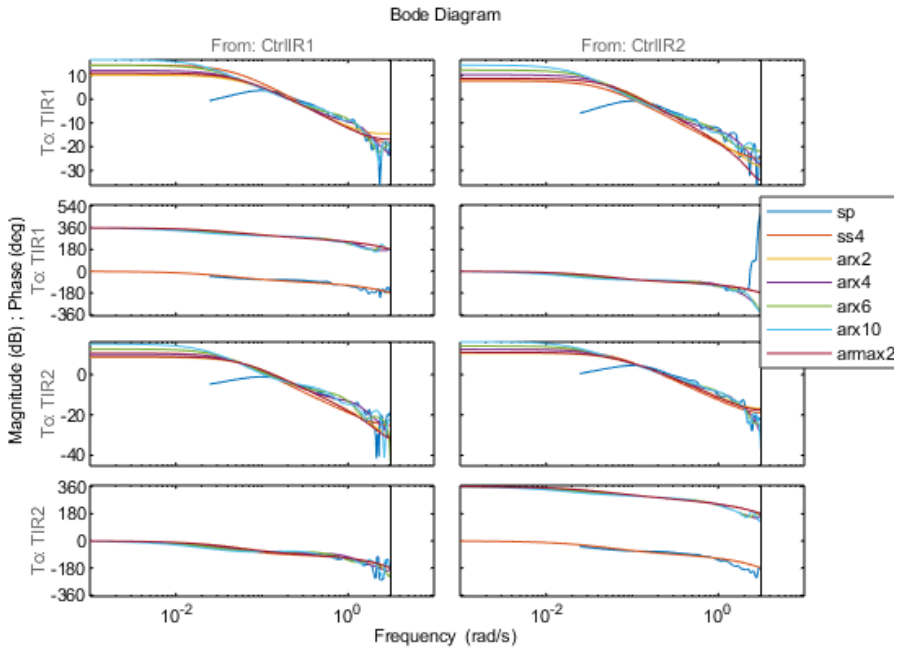


Figure 3.12: Bode plot from analysed data and estimated models for two input/output relations.

Furthermore, the models are validated by examination of the placement of their zeros and poles. Figures 3.13, 3.14 and 3.15 illustrate how they are located for two input/output combinations for the relevant models. In the figures, the circles represent zeros and the crosses represent poles. The most important observation is that all of the models have their poles within the unit circle, meaning they represent stable systems.

Many of the poles and zeros in all of the illustrated models are clustered close together. This does not have to be problematic, since the confidence intervals in most cases are quite small, however it can still raise questions about the model orders.

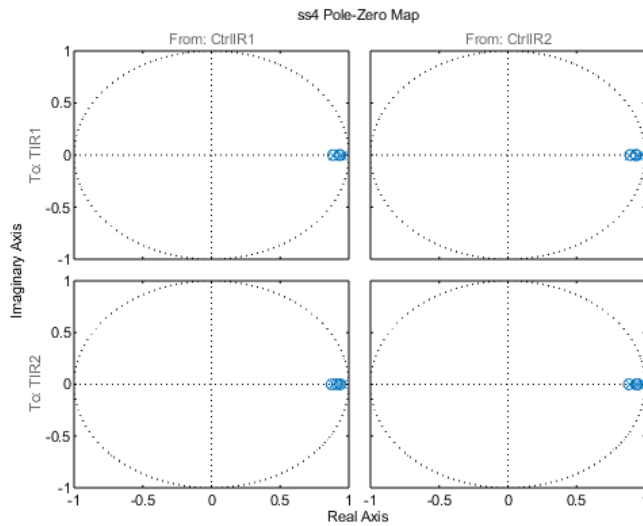


Figure 3.13: Pole-zero map of two input-output combinations for a state-space model of order 4.

Another observation is that both the ARX and ARMAX models illustrated has a zero placed outside the unit circle, indicating a non-minimum phase system. This is hard to motivate in the physical system, however the uncertainties around these zeros are typically rather large which might motivate keeping the models. In fact, the only thing the models are sure about regarding this stray zero is that it should be located somewhere on the real axis.

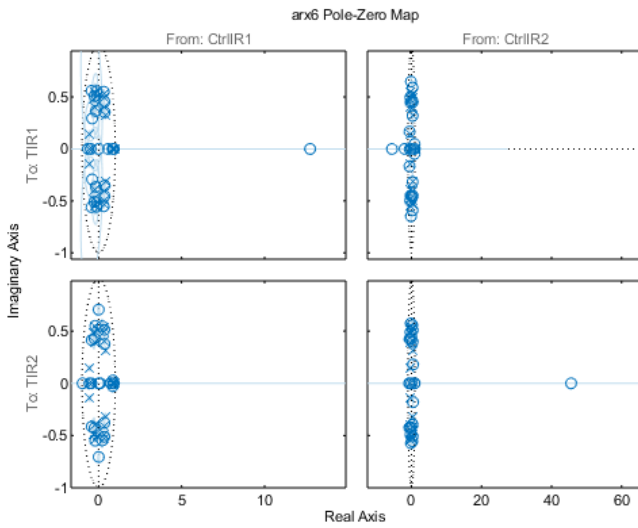


Figure 3.14: Pole-zero map of two input-output combinations for a ARX model of order 6.

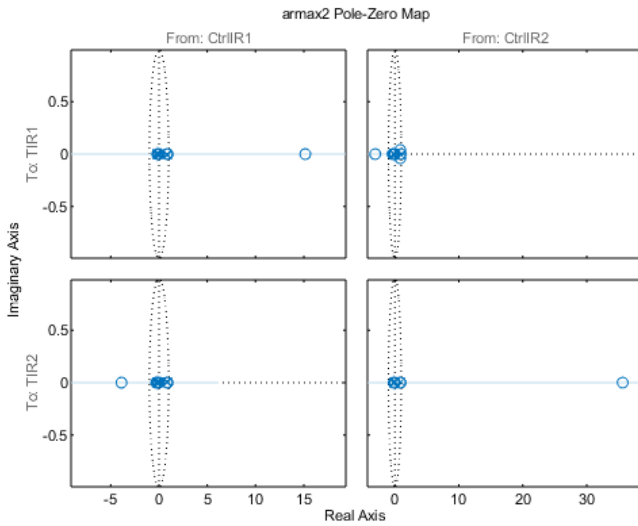


Figure 3.15: Pole-zero map of two input-output combinations for a ARMAX model of order 2.

The residuals give information on whether or not the output of the model is based on all the available input to the system. If the residuals of a model does not stay

inside its corresponding confidence interval it indicates there is unmodeled dynamics that should be included in the model.

The residuals for the state-space model can be seen in Figure 3.16. From the figure it can be seen that the residuals are fairly white indicating that most of the system dynamics are covered by the model. However, the model seems to be slightly better at modelling the cross connections in the system than the diagonal connections.

Residuals for the ARX and the ARMAX models are shown in Figure 3.17 and Figure 3.18, respectively. These residuals are very similar to the residuals from the state-space model.

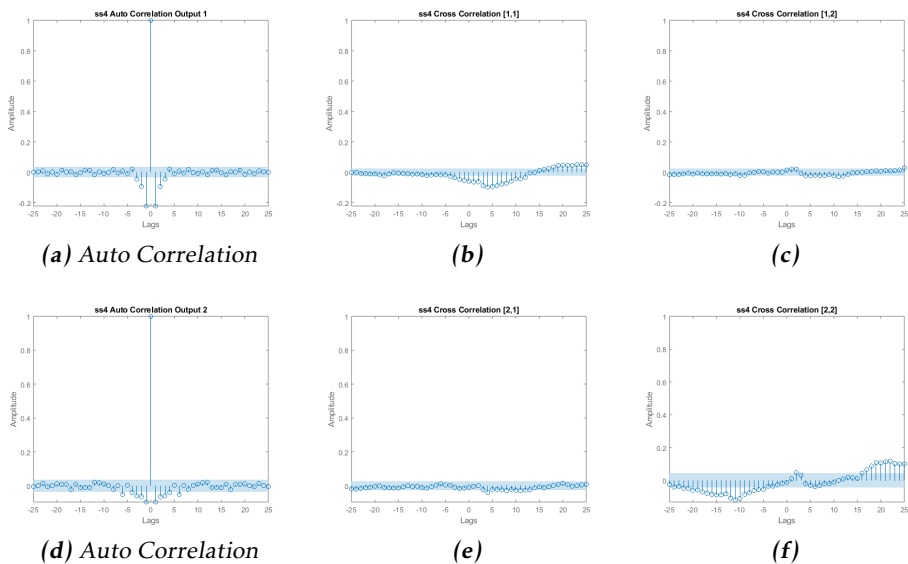


Figure 3.16: Residuals for two input-output combinations for a state-space model of order 4.

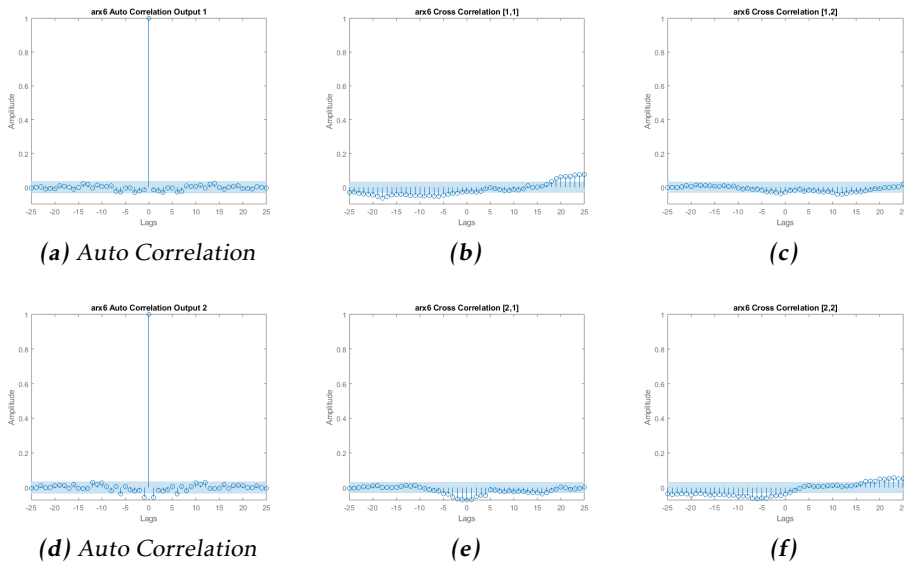


Figure 3.17: Residuals for two input-output combinations for an ARX model of order 6.

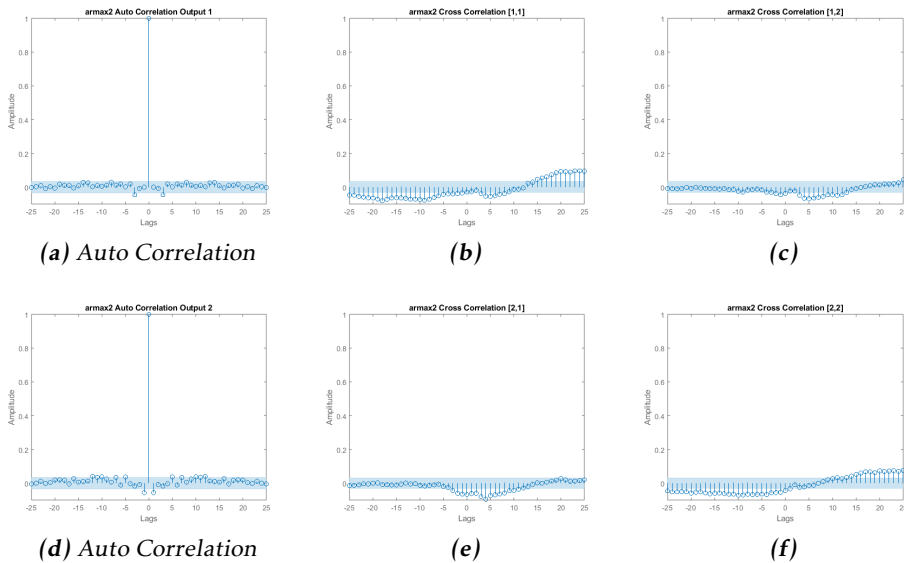


Figure 3.18: Residuals for two input-output combinations for an ARMAX model of order 2.

3.4.2 Grey-box Model Validation

Similar to the black-box model validation the output fit ratio, step response, Bode diagram, zeros and poles placement and residuals are examined to validate a grey-box model.

The model fit was presented in Section 3.3.1 and is 56%, which is considered acceptable. Figure 3.19 illustrates the step response where two areas are shown. From the step response it can clearly be seen that the diagonal elements dominate, which is logical.

Figure 3.20 shows a comparison between a spectral estimate, using Blackman-Tukey estimation, compared to the grey-box model for two of the input-output relations. From Figure 3.20 it can be concluded that the model captures the system fairly well.

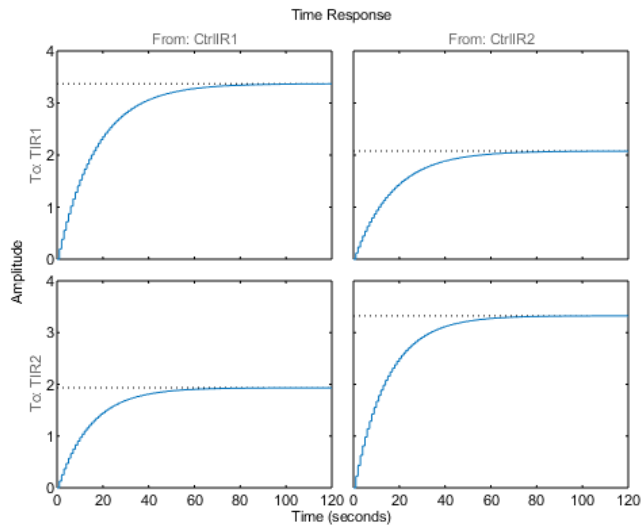


Figure 3.19: Step response for two input-output combinations for a grey box model.

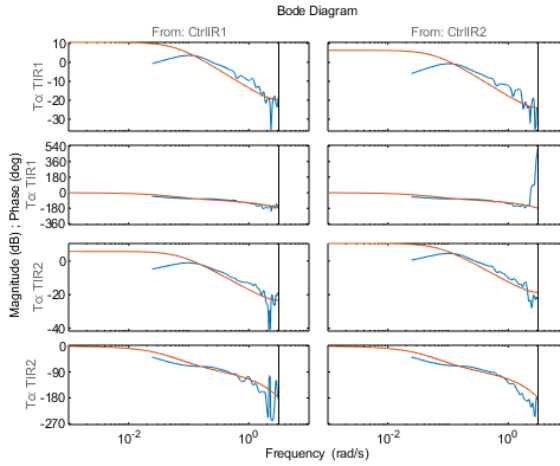


Figure 3.20: Bode plot comparing a Blackman-Tukey estimate to the grey-box model for two of the outputs.

The poles and zeros for two input/output relations can be seen in Figure 3.21. The poles and zeros are located inside the unit circle showing that the model is stable and minimum phase. All the different input-output relations have their respective poles and zeros inside the unit circle making the whole model stable and minimum phase.

The residuals indicate whether the model contains all available information. Figure 3.22 illustrates the auto-correlation function for two of the outputs and it

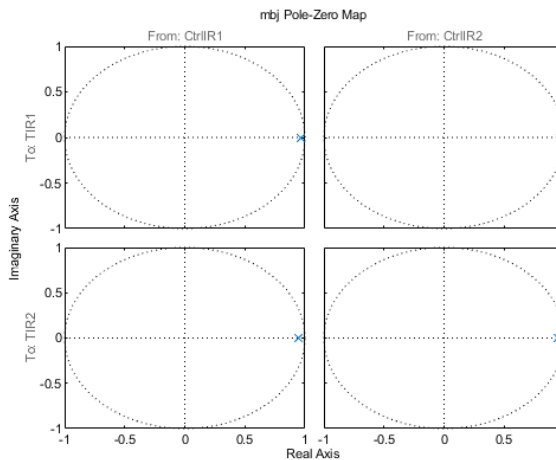


Figure 3.21: Poles and zeros for two of the input/output relation.

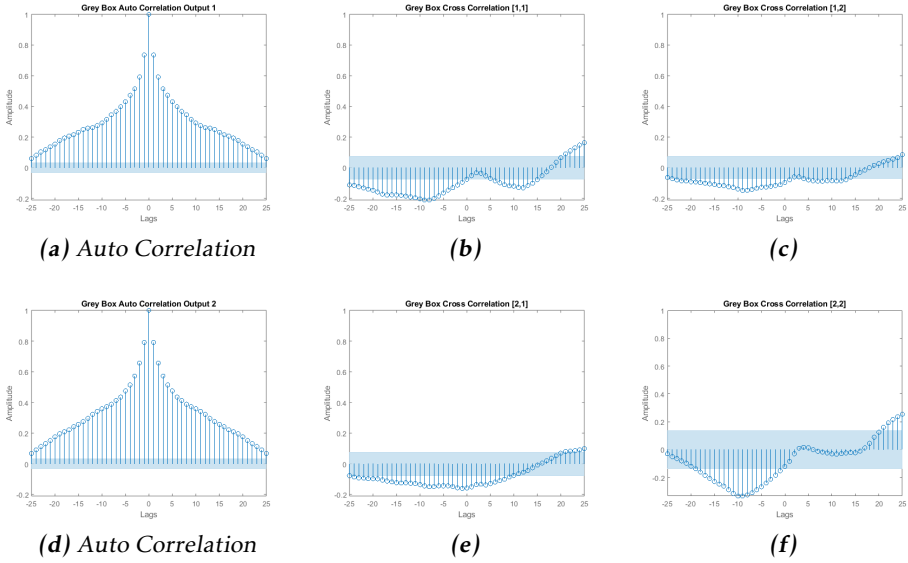


Figure 3.22: Residuals for two input-output combinations of a grey box model.

also shows the cross-correlation for two of the input-output relations. Both the auto-correlation and the cross-correlation functions indicate that the model does not explain all aspects of the data.

By estimating the grey-box models, we get estimates of the parameters that the model is built upon in Section 3.3. These parameters should correspond directly to physical properties in the real system. An arbitrarily chosen area and one of its neighbours has its parameters estimated to be as follows:

- $R_{s,i} \approx 100$
- $C_i \approx 0.15$
- $n \approx \begin{pmatrix} 0.03 & 0.02 & \cdots \\ 0.02 & 0.03 & \\ \vdots & & \ddots \end{pmatrix}$

Worth noting is that some areas of the printer have bigger fluctuations in temperature and the effect this has on the estimated parameters can be seen in the parameter values. One of the benefits of a grey-box model is that the estimated parameters correspond to real physical attributes of the system. Therefore, conclusions of where the isolation could be improved, if a heat source is damaged, etc., can be drawn.

3.5 Model Selection

The optimal model is chosen based on the model validation in Section 3.4.1 and Section 3.4.2 and will be used in the forthcoming chapters.

From Table 3.1, where the model fit averages are compared, the BJ and OE models gives considerably worse results and will therefore be discarded. The grey-box model, with an model fit of 56%, has a lower result than most of the remaining black-box models. However, since the grey-box model contains physical parameters the model is given some leeway.

From analysing step responses like those shown in Figure 3.10 and Figure 3.11, all of the initial models are excluded from consideration due to unrealistically strong cross correlations. In their place, new models with lower model orders are estimated and their step responses checked. These are the models listed in Table 3.2.

Analysing the Bode diagrams in Figure 3.12 and Figure 3.20 it is observed that all the models capture the dynamics of the system relatively well. However, the ARX model of order 6 captures the high frequency dynamic of the system best. Based on this and the fit values in Table 3.2, four models are chosen for further validation. These are the state-space model of order 4, the ARX model of order 6, the ARMAX model of order 2 and the grey-box model.

The placement of poles and zeros in Figures 3.13, 3.14, 3.15 and 3.21 does not raise any major concerns about any of the models, other than confirming that there is some uncertainty left in the models. This is best illustrated by the zeros with large confidence intervals outside the unit circle for the ARX and ARMAX models.

As for the poles and zeros placement, the residuals in Figures 3.16, 3.17 and 3.18 cannot exclude any of the black-box models from consideration since they are all rather similar. The residuals for the grey-box model shown in Figure 3.22 on the other hand, do indicate that the model does not utilise all of the information available in the data. However, the authors do not see this as a reason for exclusion, only as an indicator that the model can be developed further.

Consequently, the models that are tested further in the forthcoming chapters are the state-space₄, ARX₆, ARMAX₂ and grey-box models. The number indicate model order. Since it has a slightly higher model fit, the ARX model have a slight advantage compared to the other models.

Parts of the models are presented below, with the state-space₄ matrices described

by

$$A_{ss} = \begin{pmatrix} \ddots & & & \\ & 0.8952 & -0.000186 & \\ & 0.005608 & 0.8657 & \\ & & & \ddots \end{pmatrix} \quad B_{ss} = \begin{pmatrix} \ddots & & & \\ & 0.002616 & -0.001675 & \\ & -0.003458 & 0.005641 & \\ & & & \ddots \end{pmatrix} \quad (3.11) \quad (3.12)$$

$$C_{ss} = \begin{pmatrix} \ddots & & & \\ & 3.85 & -6.945 & \\ & -3.486 & 11.03 & \\ & & & \ddots \end{pmatrix} \quad D_{ss} = \begin{pmatrix} \ddots & & & \\ & 0 & 0 & \\ & 0 & 0 & \\ & & & \ddots \end{pmatrix} \quad (3.13) \quad (3.14)$$

and

$$K_{ss} = \begin{pmatrix} \ddots & & & \\ & 0.008907 & -0.009778 & \\ & -0.01978 & 0.03911 & \\ & & & \ddots \end{pmatrix} \quad (3.15)$$

. The ARX6 model is partly described by

$$y_{arx6}[t] = \begin{pmatrix} \ddots & & & \\ & \frac{-0.0178+0.2133z^{-1}+0.1182z^{-2}+0.04644z^{-3}-0.03555z^{-4}-0.03178z^{-5}}{1-0.6176z^{-1}-0.04991z^{-2}-0.06038z^{-3}-0.04111z^{-4}-0.006644z^{-5}-0.04602z^{-6}} & & \\ & \frac{-0.004897+0.2191z^{-1}+0.1196z^{-2}-0.0262z^{-3}-0.02419z^{-4}-0.003775z^{-5}}{-0.06944z^{-1}+0.00914z^{-2}-0.005304z^{-3}+0.02117z^{-4}+0.007697z^{-5}-0.001183z^{-6}} & & \\ & & & \ddots \end{pmatrix} u[t] \quad (3.16)$$

, the ARMAX2 model by

$$y_{armax2}[t] = \begin{pmatrix} \ddots & & & \\ & \frac{-0.01604+0.2362z^{-1}}{1-0.7651z^{-1}-0.09363z^{-2}} & & \\ & \frac{-0.007039+0.2465z^{-1}}{1-0.8324z^{-1}-0.03124z^{-2}} & & \\ & & & \ddots \end{pmatrix} u[t] + \begin{pmatrix} \ddots & & & \\ & \frac{1-0.1311z^{-1}-0.1285z^{-2}}{1-0.7651z^{-1}-0.09363z^{-2}} & & \\ & \frac{1-0.03969z^{-1}-0.06352z^{-2}}{1-0.8324z^{-1}-0.03124z^{-2}} & & \\ & & & \ddots \end{pmatrix} e[t] \quad (3.17)$$

and the grey-box model by

$$A_{greybox} = \begin{pmatrix} \ddots & & & \\ & 0.9417 & 0 & \\ & 0 & 0.9325 & \\ & & & \ddots \end{pmatrix} \quad B_{greybox} = \begin{pmatrix} \ddots & & & \\ & 0.1966 & 0.1211 & \\ & 0.1305 & 0.2245 & \\ & & & \ddots \end{pmatrix}, \quad (3.18) \quad (3.19)$$

$$C_{greybox} = \begin{pmatrix} \ddots & & & \\ & 1 & 0 & \\ & 0 & 1 & \\ & & & \ddots \end{pmatrix} \quad (3.20) \quad D_{greybox} = \begin{pmatrix} \ddots & & & \\ & 0 & 0 & \\ & 0 & 0 & \\ & & & \ddots \end{pmatrix} \quad (3.21)$$

and

$$K_{greybox} = \begin{pmatrix} \ddots & & & \\ & 0.06007 & 0 & \\ & 0 & 0.06991 & \\ & & & \ddots \end{pmatrix} \quad (3.22)$$

4

Control Design

When deciding what control strategy to choose, the intended usage is an important consideration. This chapter describes the different control designs that are used as well as the simulations based on those controllers.

4.1 Selecting Model for Control Design

In all of the proposed controllers, there is a need to use a model of the system when designing the controllers. In this section it will be described how to pick the model yielding the most desirable results. Using the models obtained in Section 3.5, a diagonal PI controller is constructed using lambda tuning [9] based on the models simulated step response. This controller, when based on the state-space4 model, is partly described by

$$F_{diagonal}(z) = \begin{pmatrix} \ddots & & & \\ & \frac{0.9626-0.9043z^{-1}}{1-z^{-1}} & 0 & \\ & 0 & \frac{0.8812-0.8244z^{-1}}{1-z^{-1}} & \\ & & & \ddots \end{pmatrix} \quad (4.1)$$

. Any model that resembles the physical system should be able to be used to simulate the effect this controller would have in a closed loop system. Simulation of a step response of the closed loop system yields the results shown in Figures 4.1, 4.2, 4.3 and 4.4. The simulations are executed in the MATLAB add-on program Simulink.

As usual, these figures only show the measurements from two of the IR sensors,

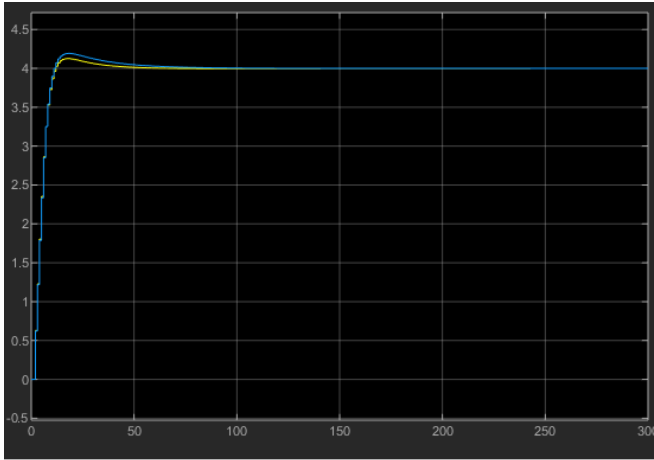


Figure 4.1: A simulated step response with two of the IR sensors using a diagonal PI controller and a state-space model of order 4.

but keep in mind that the system can have multiple measurements with significant cross correlations. The reference step is 4°C on all channels which is a reasonable temperature difference that the models should be able to handle. This also seems to be the case as illustrated in Figures 4.1, 4.2, 4.3 and 4.4. All of the black box models yield similarly satisfying results.

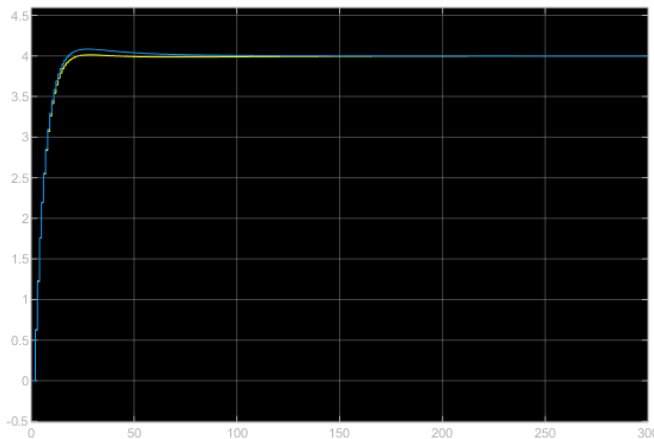


Figure 4.2: A simulated step response with two of the IR sensors using a diagonal PI controller and a ARX model of order 6.

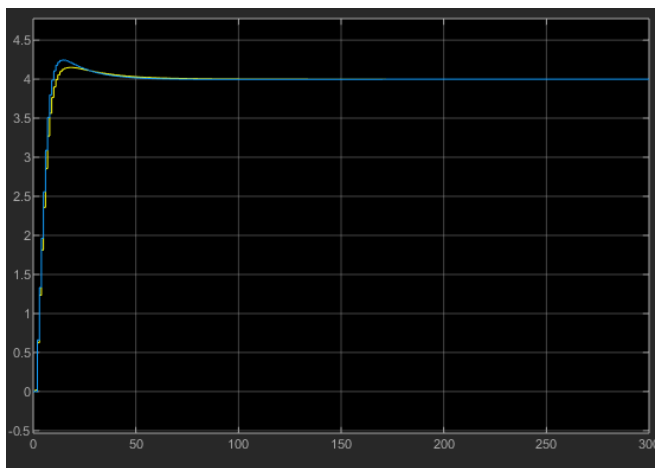


Figure 4.3: A simulated step response with two of the IR sensors using a diagonal PI controller and a ARMAX model of order 2.

Since the results of the black box models have been similar in almost all the validation tests done, there is no clear best choice. However, the state-space4 model will be the main focus in the remainder of this chapter due to its reliability to predict outputs regardless of the data set used.

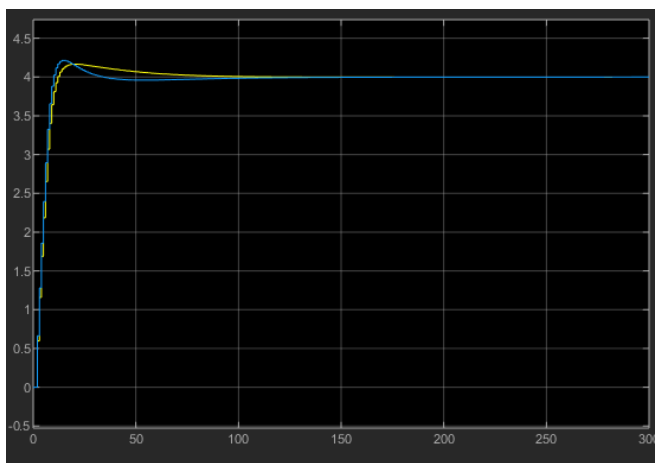


Figure 4.4: A simulated step response for two of the IR sensors using a diagonal PI controller and a grey box model.

4.2 Decoupled PI Controller

The initial attempt to improve the closed loop system behaviour was to perform decoupling of the circuit gain of the previous diagonal PI controller. The first step of this is to determine if decoupling is even necessary. This is done by performing an RGA analysis on the proposed models, which result in

$$RGA_{ss4} = \begin{pmatrix} 3.9910 & 0.4886 & \cdots \\ 1.1068 & 3.6861 & \\ \vdots & & \ddots \end{pmatrix} \quad RGA_{arx6} = \begin{pmatrix} 7.0333 & 0.5456 & \cdots \\ 1.4610 & 6.6109 & \\ \vdots & & \ddots \end{pmatrix} \quad (4.2) \quad (4.3)$$

$$RGA_{arx2} = \begin{pmatrix} 3.6142 & 0.9724 & \cdots \\ 1.1378 & 5.0513 & \\ \vdots & & \ddots \end{pmatrix} \quad RGA_{greybox} = \begin{pmatrix} 2.0413 & 0.1619 & \cdots \\ 0.0824 & 2.3239 & \\ \vdots & & \ddots \end{pmatrix} \quad (4.4) \quad (4.5)$$

. This analysis reveal a clear diagonal dependence but with significant cross-correlation for the black box models, and less cross-correlation for the grey box model. Despite the grey box models diagonality, the conclusion is drawn that the system contains significant cross-correlations.

At least for the black box models, there is clear indication that decoupling could be beneficial. Decoupling is therefore implemented on the PI controller used in Section 4.1 and tested using an identical step response simulation. The controller based on the state-space4 model is now partly described by

$$F_{decoupled}(z) = \begin{pmatrix} \ddots & & \\ \frac{1.561-1.467z^{-1}}{1-z^{-1}} & \frac{0.2333-0.2191z^{-1}}{1-z^{-1}} & \\ 0.3804-0.3559z^{-1} & \frac{1.41-1.319z^{-1}}{1-z^{-1}} & \\ & & \ddots \end{pmatrix} \quad (4.6)$$

The step response of the closed loop system for the state-space4 model is shown in Figure 4.5. The step response appears to be roughly as fast when using a decoupled controller compared to using a diagonal controller. A positive change is that the overshoot is considerably smaller.

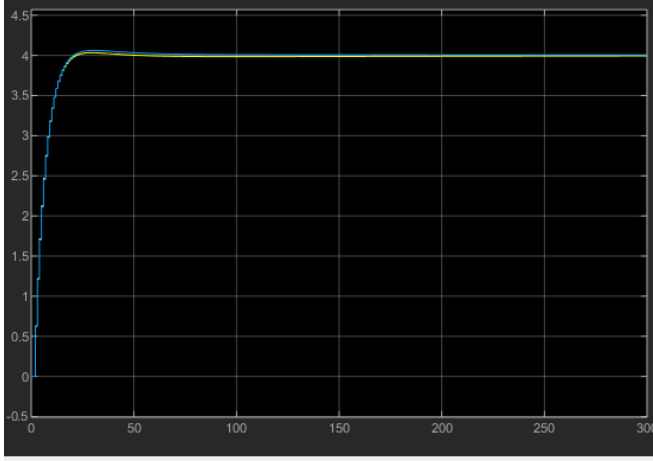


Figure 4.5: A simulated step response with two of the IR sensors using a decoupled PI controller and a state-space model of order 4.

4.3 Feed-Forward Controller

By implementing the Neutral feed-forward controller described in Section 2.2.4, the step response no longer have any overshoot as it did with the decoupled controller. However, the step time becomes slightly longer as can be seen in Figure 4.6. The largest benefit of using the feed-forward controller compared to the diagonal or decoupled PI controller seems to be that the temperatures are more synchronised during the step. This is only marginally better, but it helps in minimising the temperature gradient throughout the powder bed.

The derived controller uses the diagonal PI controller as $F(z)$, and

$$F_f(z) = \begin{pmatrix} \ddots & & & \\ & 1.622 & 0.2423 & \\ & 0.4317 & 1.6 & \\ & & & \ddots \end{pmatrix} \quad (4.7)$$

$$G_m(z) = \begin{pmatrix} \ddots & & & \\ \frac{0.1356z^{-1}-0.359z^{-2}+0.3164z^{-3}-0.09286z^{-4}}{1-3.475z^{-1}+4.521z^{-2}-2.609z^{-3}+0.5634z^{-4}} & \frac{-0.01263z^{-1}+0.03129z^{-2}-0.02526z^{-3}+0.006595z^{-4}}{1-3.475z^{-1}+4.521z^{-2}-2.609z^{-3}+0.5634z^{-4}} & & \\ \frac{0.02256z^{-1}-0.06605z^{-2}+0.06434z^{-3}-0.02085z^{-4}}{1-3.475z^{-1}+4.521z^{-2}-2.609z^{-3}+0.5634z^{-4}} & \frac{0.1381z^{-1}-0.3633z^{-2}+0.3178z^{-3}-0.09246z^{-4}}{1-3.475z^{-1}+4.521z^{-2}-2.609z^{-3}+0.5634z^{-4}} & & \\ & & \ddots & \end{pmatrix} \quad (4.8)$$

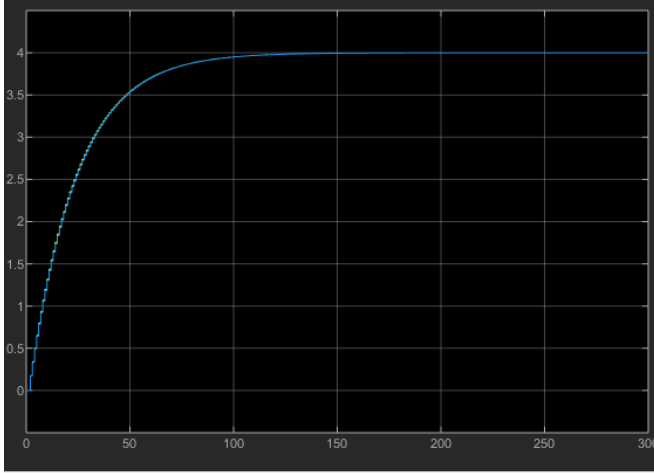


Figure 4.6: A simulated step response with two of the IR sensors using a feed forward controller and a state-space model of order 4.

4.4 IMC Controller

Implementing an IMC controller was in this thesis work not a straight forward process. As mentioned in Section 2.2.5, Q often needs to be modified since an exact inverse of G is not possible in most cases. This becomes especially clear if we would choose to use the ARX6 or ARMAX2 models which have some non-minimum phase zeros as illustrated in Figure 3.14 and Figure 3.15 respectively. When inverting G , these zeros become unstable poles which need to be remedied. The authors of [11] mention two alternatives for handling this problem, either ignore those poles or replace them with stable poles that share the same amplitude properties.

The practical implementation is kept simple by choosing the first alternative, removing the unstable parts of the system inverse. The drawback of this approach is the distorted static gain. This is compensated for by normalising Q and scaling it with the original static gain, or in other words $\tilde{Q}(z) = \frac{1/G(z)}{Q(z)} \Big|_{z=0} Q(z)$.

This process will probably introduce (if not already present) a problem with the number of poles and zeros in Q . For the IMC controller to work, Q must be proper, i.e. the number of zeros must be smaller or equal to the number of poles. This problem also has a proposed solution in [11] where $Q(s)$ is picked as $Q(s) = \frac{1}{(\lambda s + 1)^n} G^{-1}(s)$. Here, n is chosen sufficiently large in order for Q to be proper. λ is a design parameter set as the wanted bandwidth of the closed loop system. $Q(s)$ is then of course discretized into $Q(z)$. The resulting controller is partly presented

by

$$F_{IMC}(z) = \begin{pmatrix} \ddots & & & & & \\ & \frac{0.1111z^{-1}-0.1434z^{-2}-1.229z^{-3}+5.107z^{-4}-9.765z^{-5}+...}{1-7.358z^{-1}+25.67z^{-2}-56.73z^{-3}+89.29z^{-4}-106.2z^{-5}+...} & & & & \\ & -0.004907z^{-1}+0.01237z^{-2}+0.05146z^{-3}-0.2946z^{-4}+0.6517z^{-5}-0.8633z^{-6}+... & & & & \\ & & \frac{1-7.358z^{-1}+25.67z^{-2}-56.73z^{-3}+89.29z^{-4}-106.2z^{-5}+...}{1-7.358z^{-1}+25.67z^{-2}-56.73z^{-3}+89.29z^{-4}-106.2z^{-5}+...} & & & \\ & & & \ddots & & \end{pmatrix} \quad (4.9)$$

All of these considerations result in the simulated step response shown in Figure 4.7. The behaviour of the simulation should not be viewed as proof that the model will perform as good in real life. Since the simulated system and the estimated model used in the controller is the same, there will be no model error in the simulation. Still, in the simulation the reference follows as desired without oscillations and the overshoot is minimal. This can be compared with the step response of the state-space4 model in Figure 4.8 that does not have any zeros outside the unit circle. Since, the simulated model and the model that the controller is derived from is not the same some model errors will be introduced. In reality that is how the controller would function and as Figure 4.8 illustrate the controller seems to be functional.

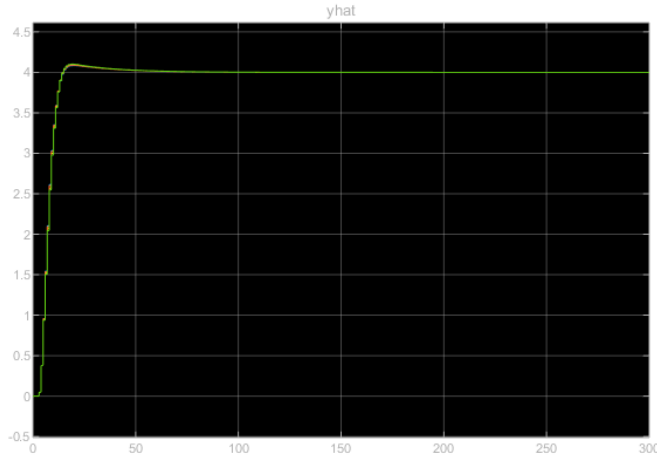


Figure 4.7: A simulated step response with two of the IR sensors using an IMC controller and a ARX model of order 6.

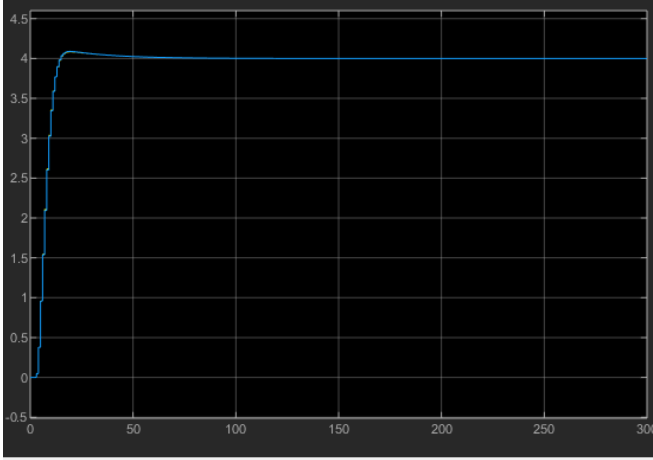


Figure 4.8: A simulated step response with two of the IR sensors using an IMC controller and a state-space model of order 4.

4.5 State Feedback Control

A state feedback controller is relatively easy and intuitive to implement. If not all states are measurable an observer is implemented to compensate for the lack of measurements. The general structure can be seen in Figure 2.4.

The controller is implemented using the LQG based functions `dlqr` and `dlqe` in Matlab. For the feedback matrix L the input is punished more than the output,

$$Q_1 = \begin{pmatrix} 1 & 0 & \cdots \\ 0 & 1 & \\ \vdots & & \ddots \end{pmatrix} \quad (4.10)$$

$$Q_2 = 1000 \cdot \begin{pmatrix} 1 & 0 & \cdots \\ 0 & 1 & \\ \vdots & & \ddots \end{pmatrix} \quad (4.11)$$

. This is to make the input to the system smaller. The observer is implemented as a Kalman filter, the R matrices are chosen as identity matrices, because no specific knowledge about the noise exists. The matrices are chosen as

$$R_1 = \begin{pmatrix} 1 & 0 & \cdots \\ 0 & 1 & \\ \vdots & & \ddots \end{pmatrix} \quad (4.12)$$

$$R_2 = \begin{pmatrix} 1 & 0 & \cdots \\ 0 & 1 & \\ \vdots & & \ddots \end{pmatrix} \quad (4.13)$$

. This yields the controller described by

$$F_y(z) = \begin{pmatrix} \ddots & & & & \\ & \frac{-2.649e-19z^{-1}+3.031e-24z^{-2}-1.156e-29z^{-3}+1.469e-35z^{-4}}{1-1.524e-05z^{-1}+8.714e-11z^{-2}-2.214e-16z^{-3}+2.11e-22z^{-4}} & & & \\ & \frac{-2.399e-19z^{-1}+2.74e-24z^{-2}-1.043e-29z^{-3}+1.324e-35z^{-4}}{1-1.524e-05z^{-1}+8.714e-11z^{-2}-2.214e-16z^{-3}+2.11e-22z^{-4}} & & & \\ & & \ddots & & \end{pmatrix} \quad (4.14)$$

$$F_r(z) = \begin{pmatrix} \ddots & & & & \\ & \frac{9.259z^{-1}+7.638z^{-2}-8.738e-05z^{-3}+3.333e-10z^{-4}-4.239e-16z^{-5}}{1-z^{-1}+1.524e-05z^{-2}-8.714e-11z^{-3}+2.214e-16z^{-4}-2.11e-22z^{-5}} & & & \\ & \frac{-0.4088z^{-1}-0.8405z^{-2}+9.394e-06z^{-3}-3.498e-11z^{-4}+4.341e-17z^{-5}}{1-z^{-1}+1.524e-05z^{-2}-8.714e-11z^{-3}+2.214e-16z^{-4}-2.11e-22z^{-5}} & & & \\ & & \ddots & & \end{pmatrix} \quad (4.15)$$

The resulting simulated step response can be seen in Figure 4.9. It is worth taking into consideration is that no disturbances are applied to the system. Hence, in reality the controller might not be as ideal as the simulation suggests.

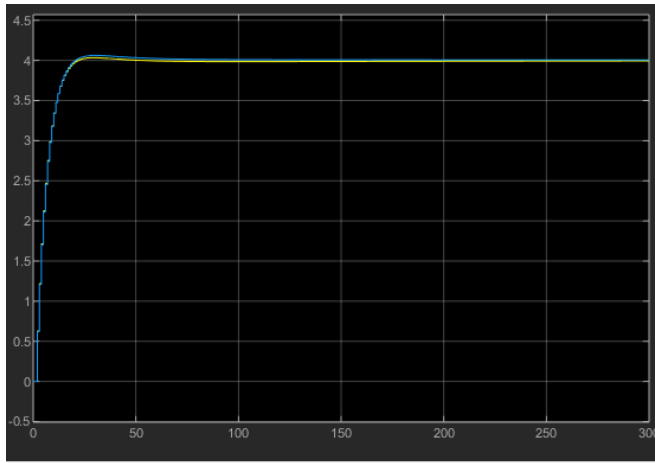


Figure 4.9: A simulated step response with two of the IR sensors using a state feedback controller and a state-space model of order 4.

5

Results

In this chapter, the results of physical tests made with the control designs are presented. Initially, the test procedure is explained, followed by the results on each of the control designs mentioned in Chapter 4.

5.1 Test Procedure

To compare the different models and control designs to each other, validation tests need to be executed. The results are presented through a value that represent how good the design performs and can be calculated for each design. Mean Square Error (MSE) will be the value that is used for evaluation of the design performance and it is calculated through

$$MSE = \frac{1}{NM} \sum_{i=1}^N \sum_{t=1}^M (y(i, t) - r(i, t))^2 \quad (5.1)$$

. In (5.1), y is the measurement at each time and for each specific sensor, r is the reference signal at the corresponding time and sensor, N is the number of different areas being measured and M is the number of measurements. The tests utilised for control design validation is a reference tracking test, a powder disturbance suppression test and a printing test.

In the reference tracking test the controller aims to bring the temperature from a set preheat temperature to a higher process temperature and then vary the reference to check the controller ability to adjust the temperature accordingly.

For the powder disturbance suppression test, powder is continuously applied to the powder bed as a disturbance while the reference temperature is fixed. This test is used due to a faulty and unreliable laser in the test machine. Furthermore, based on Figure 5.2 it can be seen that the periodical powder appliance, which lowers the temperature of the powder bed drastically, is the biggest source of disturbance to the system.

The final test is the printing test. For reproducibility reasons, it is important that a standard object is used for the print test. The default test chosen is several printouts of standard tensile test specimens, normally used for the standardised Tensile Test [7]. Figure 5.1 illustrates the default test.

It is worth considering that the default test is a relatively quick printout since not that many layers are needed for printing the objects. Hence, the test will not capture what happens when a thick powder bed has been applied and the whole system has been running for a long time. The entire print test procedure can be described by the following:

- Start the 3D printer. Heat the printer to just below working temperature. Add powder, as would be done in normal printing.
- Print the objects illustrated in Figure 5.1.
- Calculate the MSE from the logged data.

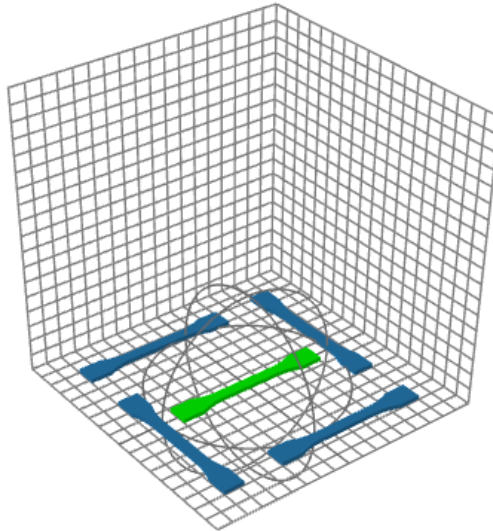


Figure 5.1: Default printout for disturbance suppression test.

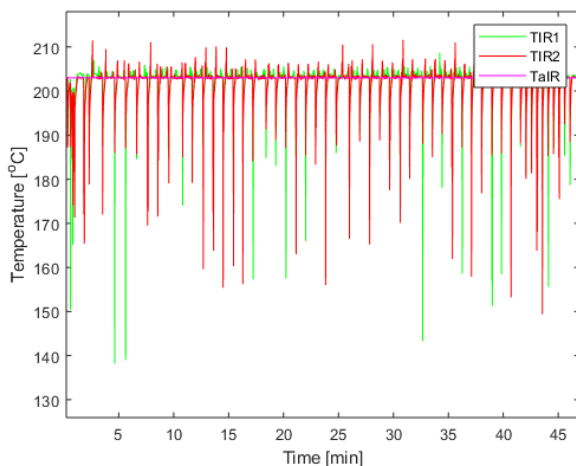


Figure 5.2: Measured temperatures from two optical sensors during a print using the original, untuned PI controller.

To have a reference point to compare the new solutions with, tests of the original solution implemented in the rebuilt 3D printer have been executed. With this, a relative improvement of the control design can be determined. As mentioned previously, the original solution is a diagonal PI controller.

The ARX model is not used for the tests written about in this report. This is because the tests are time consuming to execute and the ARX model unfortunately did not indicate good result during the initial testing in the physical system.

5.2 Reference Tracking

Controllers tested using three different models were used when testing in order to have a larger set of results to compare. The calculated MSE are shown in Table 5.1.

Table 5.1: MSE results from reference tracking using different controllers

Controller		MSE result		
Original solution		0.7338		
	Tuning model:	ss4	grey-box	armax2
Diagonal PI		0.6067	0.5871	0.6630
Decoupled PI		0.7822	0.7393	0.7086
Feed forward		0.6434	0.6392	0.7350
IMC		4.1954	1.7501	4.8535
State feedback		1.8355	1.6295	3.9235

The model that in general seems to yield the controllers that best capture the real system is the grey-box model. The controller that in general seem to give the best result is the Diagonal PI controller. The best result is given by the Diagonal PI controller based on the grey-box model and the tracking performance is shown in Figure 5.3.

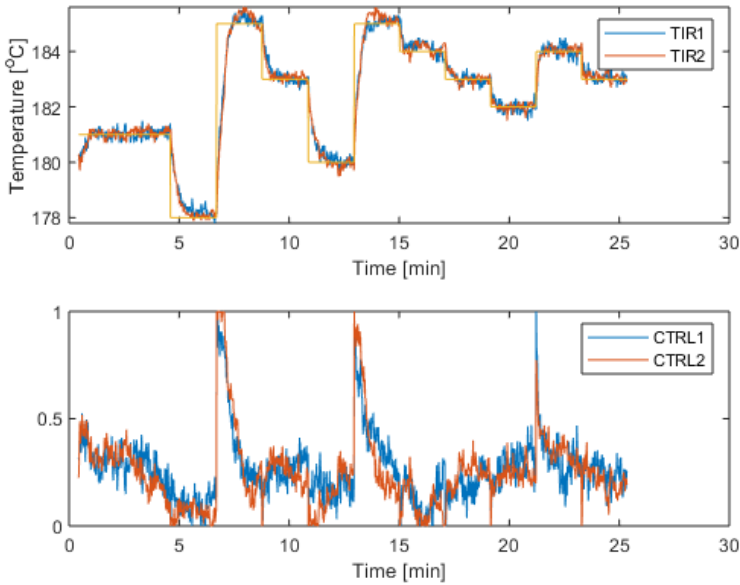


Figure 5.3: Input and output signals during a reference tracking test using a PI controller trimmed from a grey box model.

5.3 Powder Disturbance Suppression

The results from the powder disturbance suppression test can be seen in Table 5.2 where, based on the results in Table 5.1, only the relevant tests are executed. From the results it can be observed that the original solution has the best result. Why this is the case will be thoroughly discussed in Chapter 6. However, when adding a simple fuzzy control structure the results drastically improve for the diagonal PI controller based on the grey-box model. The resulting MSE is 23.2247. Hence, the new controller performs slightly better than the original controller.

The new controller is justified because when new powder is added the controller is removed from its linearisation interval. The saturation technique implemented

is described by:

$$\begin{cases} u(i) = 0, & e(i) > 2 \\ u(i) = 1, & e(i) < -2 \\ u(i) = F_{diag}(e(i)), & |e(i)| < 2 \end{cases} \quad (5.2)$$

Table 5.2: MSE results from powder disturbance suppression using different controllers

Controller		MSE result		
Original solution		25.6145		
	Model:	ss4	grey-box	armax2
Diagonal PI		46.3138	42.0089	51.0697
Decoupled PI		-	40.0768	-
Feed forward		-	-	-
IMC		-	40.4855	-
State feedback		-	32.7185	-

Figure 5.4 shows the resulting output for two areas when the target temperature is 203° and the diagonal PI controller based on the grey-box model is used in combination with the fuzzy control structure previously described. During the powder appliance, an additional measurement disturbance is introduced that does not affect the temperature of the powder bed. This is due to the mechanics of the powder appliance where the powder bed is temporarily obscured from the sensors. Hence, Figure 5.4 contains irregular temperature spikes. These spikes are removed when the MSE is calculated by simply removing any data points with a temperature below 180°C.

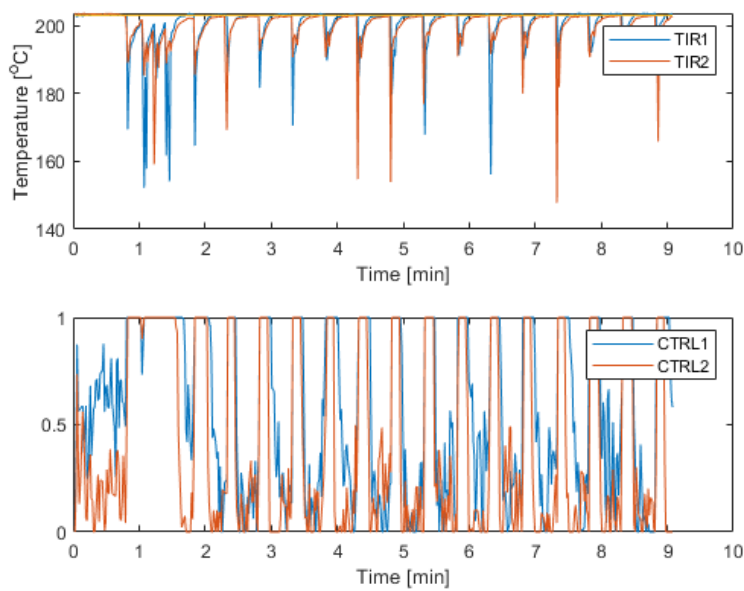


Figure 5.4: Input and output signals during a powder disturbance suppression test using a PI controller trimmed from a grey box model with a min-max fuzzy control implementation.

5.4 Print Disturbance Suppression

The previously described default test was printed for two controllers. The original solution and the fuzzy control structure with the tuned PI controller based on the grey-box model. The resulting MSE is presented in Table 5.3. It can be observed that the new controller has a better performance than the original solution.

Table 5.3: MSE results from print disturbance suppression using different controllers.

Controller	Original Solution	Tuned PI based on grey-box
MSE	22.3480	19.8853

Figure 5.5 illustrates the output and the corresponding input signals for two of the areas for the best performing controller.

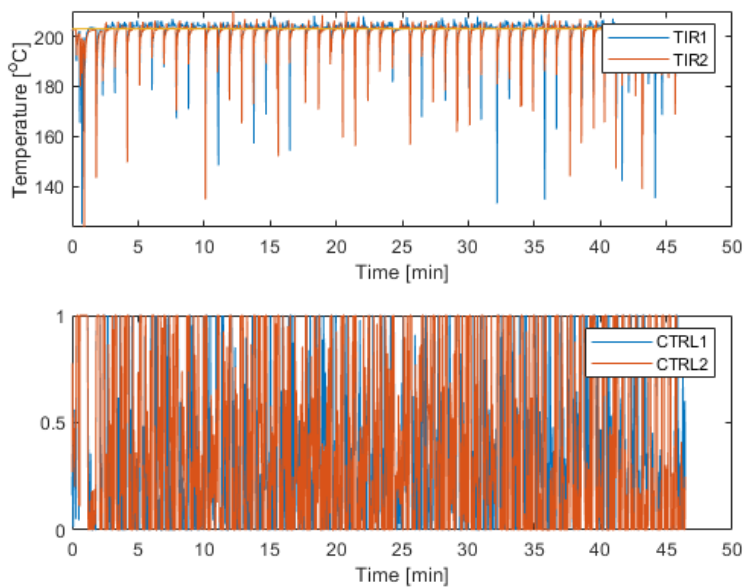


Figure 5.5: Two input and output signals during a default print test using a PI controller trimmed from a grey box model with a fuzzy control implementation.

6

Discussion

In this chapter the results above are discussed in an attempt to draw conclusions on the system and the process leading up to the results.

6.1 Models

The different black-box and grey-box models used in the work will be discussed in this section. The discussion will be based on the results presented in Chapter 5 and the derived models in Chapter 3.

6.1.1 Black-box Models

The black-box models that are more closely examined in this work are the state-space, ARX and ARMAX models. Among the black-box models, the state-space model of order 4 seems to give the best results, see Table 5.1 and Table 5.2. This is not surprising since according to L. Ljung in [15], the state-space structure is the model structure that generally manages to capture the dynamics of advanced MIMO systems well.

The higher order black-box models do not create a system with dominant diagonal elements, which is counter-intuitive to the real system. By proximity, the heating element that is closest to its corresponding area should be the dominant element. Hence, one might wonder why all the higher order models seem to create incorrect models. The authors assumes this mostly has to do with overfitting to the noise that occurred during the data measurement or failing to linearise some non-linear phenomena.

Lower order models do have dominant diagonal elements. Since the lower order

models have less possibility to model advanced dynamics or noise they are not as likely to overfit the data.

In Section 3.4.1, the state-space4 model and the ARMAX2 model are validated. Comparing Figure 3.13 and Figure 3.15 the difference between the two models' poles and zeros can be seen. The zeros for the ARMAX2 model are outside the unit circle indicating the model describes a non-minimum phase system. This has practical implications when using the inverse of the model, which for example is the case when creating an IMC controller, since this introduces an unstable pole that needs to be compensated for. This compensation often leads to changed behaviour compared to the desired one.

Furthermore, in Figure 3.16 and Figure 3.18 the residuals can be compared. The residuals for the state-space4 indicates that the cross-correlations for the diagonal elements are not being captured very well. Since the diagonal elements are most important for most controllers this is not ideal. Similarly, these effects can be seen for the ARMAX2 model. Consequently, there is still a lot of room for improvements of the models.

6.1.2 Grey-box Models

The grey-box model yields generally good results. The model is based on physical reasoning, which usually results in good models. However, since a 3D printer consists of advanced thermodynamical system dynamics with relatively undefined thermal barriers, the model was presumed to yield poor results. However, when testing controllers based on the grey-box model, they yielded good performance, see Table 5.1 and Table 5.2.

Several grey-box models were created, see Appendix A, where different approaches were examined. The model yielding the best result, presented in Section 3.3, assumes that no heat transfer occurs between different areas and where the temperature of each area is modelled as a state. Furthermore, the model assumes that each heating element affects all of the different powder bed areas. This can be compared to the alternative model derived in Section A.1 where heat transfer occur between areas and each heating element only affects its own corresponding area. Since the model presented in Section 3.3 yields better results than the alternative model, presented in Section A.1, it can be concluded that the physical assumptions for the model in Section 3.3 better describe the real system.

The main advantage of using a grey-box model over a black-box model is that the estimated parameters correspond to some real physical properties. Hence, information about the real physical system can be extracted from this estimated model. Section 3.4.2 contains some of the estimated values for the model. From the values it can clearly be seen that the energy transfer from a heating element decreases as the distance increases. Furthermore, the conductive element repre-

sents how well certain areas can retain heat and hence how well isolated they are. From the estimated parameters, clear indications of how to change the printer to further improve the dynamics are given.

In Section 3.4.2 the derived grey-box model is validated. The residuals in Figure 3.22 is the only part of the validation process of the model that stands out, giving rather poor results. Neither the auto-correlation graphs nor the cross-correlation graphs are within their respective confidence interval. However, since the grey-box model is forced into a specific state-space structure based on the physical modelling, the model does not minimise the prediction errors as well as the black box models do since the black box models are constructed with this minimisation in mind. Consequently, these results are to be expected. From Table 5.1 and Table 5.2 we see that the controller based on the grey-box model yields the best results. Therefore, one can conclude that the physical structure used in the grey-box model must be relatively correct.

6.2 Controllers

This section contains a discussion of the choices and motivations regarding the different controllers examined, as well as the impact this has on the results presented in Chapter 5.

6.2.1 Original Solution

The PI controller originally implemented in the machines was not tuned but yields surprisingly good results, at least compared to the controllers tested in this thesis work. This is either due to luck when the parameters were chosen, or due to some very basic system dynamics where the controller mostly has to suppress non-trivial amounts of noise. The controller is very aggressive which induces overshoot in some instances. However, due to the nature of the system, these are not as large as they would have been in a completely linear system. Instead, this aggressiveness is what makes the controller successful since it is able to quickly compensate for sudden temperature drops.

6.2.2 PI Controllers

The designed controllers yielding the best results in Chapter 5, are also the least advanced controllers, namely the tuned diagonal and decoupled PI controllers. Both are tuned from a generated step response derived from a model of the system, so the good results are somewhat expected. The only manual tuning made is within the range of Lambda tuning, where T_c is the design parameter. Unsurprisingly, this yields similar or better results compared to the untuned PI controller.

The tuned controllers are by nature not as aggressive as the original solution. This becomes advantageous when the model is only suppressing the normal temperature fluctuation. However, when powder appliance disturbances are introduced to the system, the original solution outperforms the tuned controller. Since the

tuned controller is not as aggressive and due to how the MSE value is calculated this is only natural. The tuned PI controller is enhanced by the introduction of a simple saturation technique, see Section 5.3. When the controller is too far outside its linearisation interval either minimum or maximum power is applied as the corresponding output signal. Once the controller is in an acceptable range of the linearisation value the tuned PI controller is then once again activated. The justification for this method is that the disturbances that the powder appliance introduce are exciting the non-linear features of the system and proportionally significant that the controller based on the linear model can not be expected to handle such large non-linearities.

The reason for the apparent success of the PI controllers compared to the IMC and state feedback controllers is not entirely certain, however the authors have two possible explanations. The first is the limited use of the estimated model. As discussed earlier, there is probably non-linear dynamics present in the system that is not captured by the models. This error has a lesser impact on the controller if the model is only used to estimate a step response, compared to if the model is an intricate part of the controller design as in the IMC controller or state feedback controller. The second possible explanation is one of implementation. This is further discussed below in Section 6.2.4.

6.2.3 The Use of Feed Forward Control

The problem that is solved during this thesis work makes it possible to exclude some controller designs from the beginning. For instance, since the printing procedure does not change the reference temperature, the most important ability for a controller becomes noise suppression. This is handled by some sort of feedback, hence the feed forward controller becomes unnecessary since it almost exclusively utilises the feedback part of the design.

However, the feed-forward design is not entirely without its uses. One possible use is during a melting test, where the temperature is increased incrementally with small steps. This can of course be solved using a pure feedback controller, but the feed-forward part should help speed up the process. Unfortunately, the results point to a slightly worse behaviour compared to a pure PI controller.

6.2.4 Practical Implementation and Linearisations

Controllers do not always behave the same in practice as they do in theory. This becomes clear when inspecting the results in Chapter 5, and it has several possible explanations.

One explanation is that the system is a lot more non-linear than first suspected. This would explain why the more advanced controllers, that rely more heavily on a model of the system, perform worse than the simple PI controllers. For example, the feed-forward controller should yield better results in the reference tracking test than a PI controller, but it does not. This is most likely due to the F_f ma-

trix that is a static gain matrix based on a linear model. The model is linearised around the operating temperature, and an attempt was made to compensate for this in the F_f matrix by shifting the reference signal such that the operating temperature corresponds to zero in the input to F_f . This results in controllers that mostly do as they should, only not as good as desired and expected.

Similar linear shifts in signals are made in the state feedback controller where the results are also questionable. In fact, these compensations are needed in every case where a control block has an input signal that is not some type of error or difference signal. The authors realise that this is ad hoc solutions of sort, but see no better solution. Unfortunately the result is controllers that does not quite meet the requirements.

6.2.5 IMC Controller Revealing Interesting Dynamics

One of the more interesting observations made concerns the input signals generated from the IMC controller during the reference tracking test. As can be seen in Figure 6.1 the control signals are very different from each other. In fact, it seems that the controller (almost) manages to track a reference using only half of the available input signals in each instant. The results are not as good as many of the other controllers, but that is most likely due to the non-linearities as described above. Still, this result suggests that with sufficient knowledge about the system, a controller can adjust the temperature in unconventional and unintuitive ways.

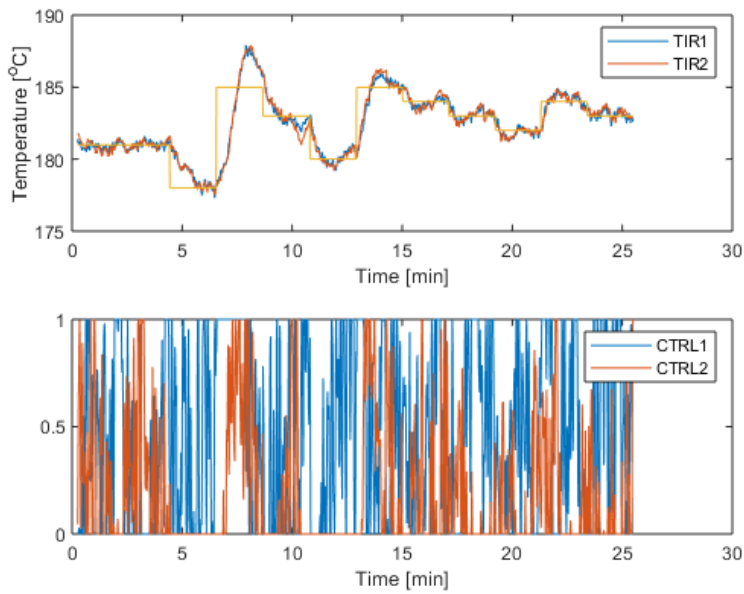


Figure 6.1: Input and output signals during a reference tracking test using a IMC controller based on a grey box model.

7

Conclusions

This chapter aims to summarise and concretise any conclusions drawn in this thesis work.

7.1 Models

The main conclusion one can draw regarding the models is that the simpler models, such as the grey-box and ARX models, outperform the more complex models, such as the BJ and OE models. Whether this is because of simple system dynamics or non-linear discrepancies is hard to say, but the results are quite clear. The higher order models either generate unintuitive step responses, or become unreliable.

The grey box model seems to yield the most promising result, indicating this to be the preferred path forward. It is also the model with the largest development potential since it can easily be modified as Appendix A gives examples of.

In conclusion, an answer to the original question:

- Can the system be described by a linear parametric model?

can be given as:

- Yes, the system can be described by a linear parametric model in a small temperature interval. If the temperature deviates too far from the linearisation point, the model becomes invalid.

7.2 Controllers

Similarly to the models, the controllers yielding the best results are often the simple designs such as the original PI controller and the tuned diagonal PI controller, where the latter has a slight advantage.

The design of the printer appears to be sub-optimal since the IMC controller more or less is able to control the heat using a subset of available control elements. The authors believe a more thermodynamically stable system has to be built, i.e. better isolation, more control signals, etc, if further improvements of the temperature control is to be achieved in the printer. Note that testing was executed on an older printer. Hence, the newer versions might have a more stable thermodynamical picture. The current control signals are only able to affect the temperature to a certain degree, which is not good enough to create a controller that results in a small *MSE*.

The linear controllers examined seem to be good at controlling the temperature in small intervals, but have trouble controlling the temperature over larger temperature intervals. During a print, when both powder and laser disturbances are in effect, the temperature fluctuates over a large temperature interval as seen in Figure 5.5. Hence, it can be concluded that either several linear controllers or some non-linear controller is needed to more accurately control the temperature over large temperature spans.

In conclusion, the original questions surrounding the controller:

- Can the temperature of the 3D printer be controlled using a controller based on a MIMO model?
- Does the use of the aforementioned controller result in better temperature control than the original solution does under realistic conditions as described in Section 1.2?

can be answered:

- The answer to both questions is yes. The PI controller based on the grey-box model performs better than the original solution under realistic conditions. However, the results are not that much better than the original solution. The reason for this is that the disturbances during a print create non-linearities that the controller based on the linear model are not tuned to handle.

8

Future Work

This chapter aims to give examples and recommendations on how to continue and expand the work done in this thesis project. Different observations surrounding the dynamics of the printer are also discussed.

8.1 Non-linear Models

One of the clearest conclusions drawn regarding the system is that it is significantly non-linear. Even though models linearised around an operating temperature work for fitting PID controllers, they are clearly not good enough for designing more complex controllers. However, non-linear models might be and according to the authors the logical next step. With these models, the model dependent controllers might yield better results.

8.2 Heating Elements and Sensors

The number of control and measurement signals clearly constitutes a limitation in how well the system can be modelled and controlled. By increasing the amount of controllable heating elements or the amount of sensors, the controller gets more information and the temperature can be better controlled.

More points of measurements is especially interesting for the modelling. In [21], 101 measurement points are used for modelling of an oven. Thermodynamically, an oven and a PBF printer are relatively similar. Hence, this indicates that more sensors would be beneficial for the modelling.

8.3 Modelling of Laser and Powder Appliance

Since the demand for PBF printers has increased the last couple of years, more studies that try to describe the effect of the laser on the powder bed have been made [3, 5, 6, 8, 22]. Therefore, when the heat for the whole system is modelled the laser effect could be taken into consideration.

Furthermore, when the system is active it is causal. The different parts that are about to be printed are known. Hence, when and where the laser will be active is known. This information could be integrated with the temperature control of the whole system if the laser effect is shown to affect the temperature in a significant way.

Similarly as with the modelling of the laser, the new layer of powder that is sequentially applied throughout a print can be modelled. When a new layer is applied, the temperature of the layer needs to be increased to the correct value. Since it is known when the layer will be applied the heating element could for example start heating beforehand. This could be a good idea since the system is relatively slow.

The different properties that a specific powder has are also something worth taking into consideration. Since the foremost purpose of the printer is to create products out of the powder, it is essential that the powder is modelled properly. In PBF printing several different types of powders are used that have different properties [1, 4, 12, 23–25].

8.4 Alternative Control Strategies

With the knowledge obtained in this thesis work, one can design controllers that are better equipped to handle the non-linearities and large disturbances of the system. Strategies of particular interest include feed-forward from disturbance and a closer look at fuzzy control. Examples of how a fuzzy control structure can be implemented for temperature control can be found in [9] and [2]. One might also want to examine the possibility of having several linear controllers in different temperature intervals where each is optimised for their respective interval. This is usually solved by having a look-up table of parameters for the controller.

8.5 Better Isolation of the Printer

The printer used for the testing was an older model with a thermal system that was not as developed as the newer printer models. This results in more disturbances. When more uneven and irrational disturbances occur, the advanced controllers become incorrect and perform poorly. This might be why the best result

was from a tuned PI controller. If a new printer is used that has better isolation and a more stable thermal system, more advanced controllers might outperform the PI controller.

Appendix

A

Grey-box Models

This appendix describe different ways to design grey box models of the examined system. An alternative model and a extended model of the whole system are presented.

A.1 Alternative Model

The alternative model is based on the system being described using a thermal model illustrated in Figure A.1. The powder bed being modelled is, as previously mentioned, divided into N areas. One area, i , is connected to another area, j . The connection between these areas is modelled as a thermal resistance. Each modelled area has a corresponding controllable heater, a heat storage modelled as a capacitor, a heat flow connected to the area and a thermal resistance connected to the surrounding temperature.

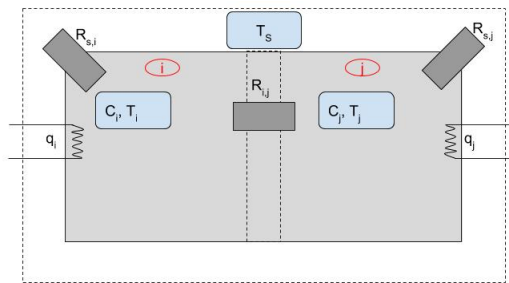


Figure A.1: Model of the thermal system with N different locations being modelled.

From Figure A.1 the bond graph in Figure A.2 is constructed.

From Figure A.2 and (2.4b), one can derive

$$q_i = \frac{1}{R_{s,i}}(T_s - T_i) \quad (\text{A.1})$$

,

$$q_{i,j} = \frac{1}{R_{i,j}}(T_i - T_j) \quad (\text{A.2})$$

and

$$C_i \dot{T}_i = q_i + q_{f,i} - \sum_{j \neq i} q_{i,j} \quad (\text{A.3})$$

. Equation (A.1) model the heat flow between a specified area i and its surrounding temperature T_s where, $i \in [1, \dots, N]$. The heat flow between the different areas is modelled using (A.2), where $i \in [1, \dots, N]$, $j \in [1, \dots, N]$ and $j \neq i$. The general flow and temperature difference for an area can be modelled by (A.3).

Inserting (A.1) and (A.2) into (A.3) yields

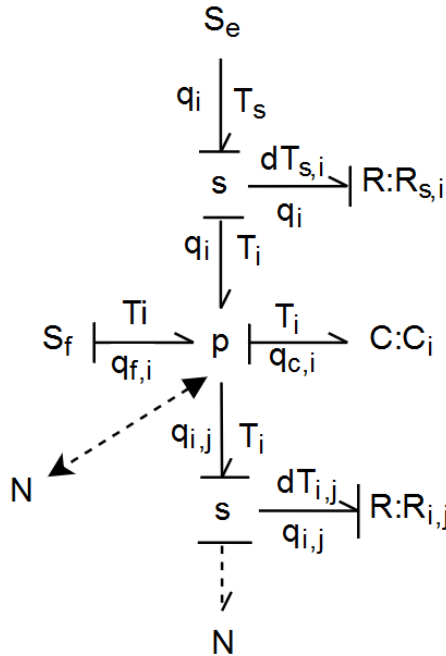


Figure A.2: Bond graph structure for N different areas being modelled.

$$\dot{T}_i = \frac{1}{C_i} \left(\frac{1}{R_{s,i}} (T_s - T_i) + q_{f,i} - \sum_{j \neq i} \frac{1}{R_{i,j}} (T_i - T_j) \right) \quad (\text{A.4})$$

. This can be extended into a state-space model. Choosing $x = [T_1, \dots, T_N]^T$, $u = [q_{f,1}, \dots, q_{f,N}]^T$ and $w = T_s$ result in

$$\dot{x}_i = \frac{-1}{C_i} \left(\frac{1}{R_{s,i}} + \sum_{j \neq i} \frac{1}{R_{i,j}} \right) x_i + \sum_{j \neq i} \frac{1}{C_i R_{i,j}} x_j + \frac{1}{C_i} u_i + \frac{1}{C_i R_{s,i}} w \quad (\text{A.5})$$

. When writing the state-space model on standard form,

$$\begin{aligned} \dot{x} &= Ax + Bu + Nw \\ y &= Cx \end{aligned} \quad (\text{A.6})$$

, the A, B, C and N matrices are given by

$$A = \begin{pmatrix} \frac{-1}{C_1} \left(\frac{1}{R_{s,1}} + \sum_{j \neq 1} \frac{1}{R_{1,j}} \right) & \frac{1}{C_1 R_{1,2}} & \cdots & \frac{1}{C_1 R_{1,N}} \\ \frac{1}{C_2 R_{1,2}} & \frac{-1}{C_2} \left(\frac{1}{R_{s,2}} + \sum_{j \neq 2} \frac{1}{R_{2,j}} \right) & & \frac{1}{C_2 R_{2,N}} \\ \vdots & & \ddots & \vdots \\ \frac{1}{C_N R_{1,N}} & \frac{1}{C_N R_{2,N}} & \cdots & \frac{-1}{C_N} \left(\frac{1}{R_{s,N}} + \sum_{j \neq N} \frac{1}{R_{N,j}} \right) \end{pmatrix} \quad (\text{A.7})$$

$$B = \begin{pmatrix} \frac{1}{C_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{C_2} & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{C_N} \end{pmatrix} \quad (\text{A.8})$$

$$N = \begin{pmatrix} \frac{1}{C_1 R_{s,1}} & 0 & \cdots & 0 \\ 0 & \frac{1}{C_2 R_{s,2}} & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{C_N R_{s,N}} \end{pmatrix} \quad (\text{A.9})$$

$$C = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \quad (\text{A.10})$$

A.2 Extended Model of Complete Thermal System

The whole system that is described in Section 1.1 can be incorporated into a grey-box model. Figure A.3 describes the thermodynamical situation of the system. The powder bed temperature is now only modelled as a single temperature. The calculations deriving a state-space model of the whole system are presented.

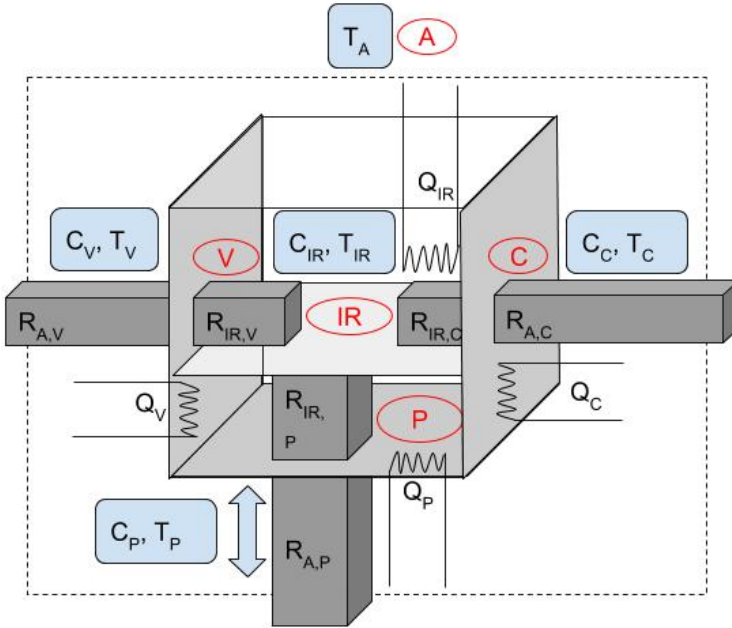


Figure A.3: Thermodynamic description of the whole system.

The different areas being modelled are:

- IR: Infra Red. The powder bed area where the temperature is controlled by an optical heating system. Corresponds to area 0 in Section 1.1.
- C: Container. The container that contains the powder, the temperature is controlled by conductive heating elements. Corresponds to area 3 in Section 1.1.
- V: Volume. Heats the deposited powder from the sides, the temperature is controlled by conductive heating elements. Corresponds to area 1 in Section 1.1.
- P: Plate. Heats the deposited powder from the bottom, the temperature is controlled by conductive heating elements. Corresponds to area 2 in Section 1.1.
- A: Ambient. The ambient temperature outside of the printer.

From Figure A.3 a bond graph can be created. The bond graph can be seen in Figure A.4.

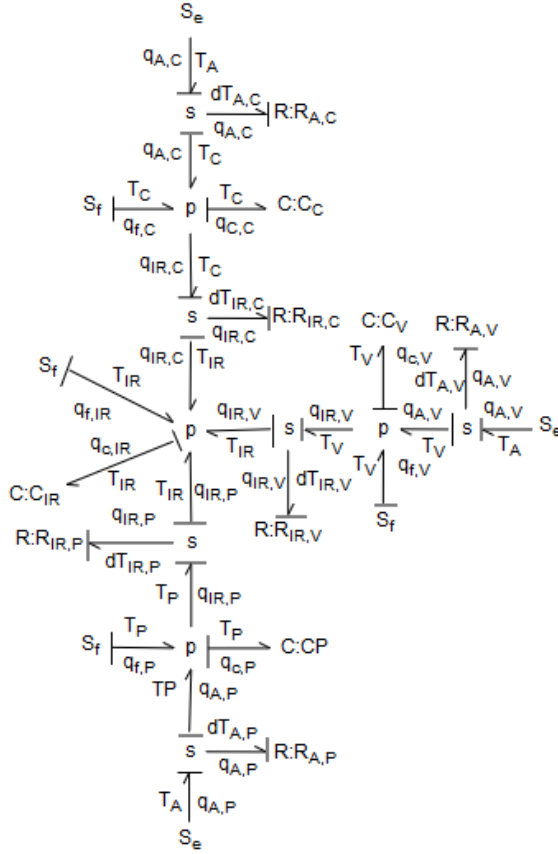


Figure A.4: Bond graph for the whole thermal system.

From the bond graph in Figure A.4 we can see that the system is casual. The temperature flow between the different areas can be modelled by

$$C_{IR}\dot{T}_{IR} = q_{f,IR} + q_{IR,C} + q_{IR,V} + q_{IR,P} \quad (\text{A.11})$$

$$C_C\dot{T}_C = q_{f,C} + q_{A,C} - q_{IR,C} \quad (\text{A.12})$$

$$C_V\dot{T}_V = q_{f,V} + q_{A,V} - q_{IR,V} \quad (\text{A.13})$$

and

$$C_P\dot{T}_P = q_{f,P} + q_{A,P} - q_{IR,P} \quad (\text{A.14})$$

. When the corresponding flows, q , are inserted the flows can be extended into

$$C_{IR}\dot{T}_{IR} = q_{f,IR} + \frac{1}{R_{IR,C}}(T_C - T_{IR}) + \frac{1}{R_{IR,V}}(T_V - T_{IR}) + \frac{1}{R_{IR,P}}(T_P - T_{IR}) \quad (A.15)$$

$$C_C\dot{T}_C = q_{f,C} + \frac{1}{R_{A,C}}(T_A - T_C) - \frac{1}{R_{IR,C}}(T_C - T_{IR}) \quad (A.16)$$

$$C_V\dot{T}_V = q_{f,V} + \frac{1}{R_{A,V}}(T_A - T_V) - \frac{1}{R_{IR,V}}(T_V - T_{IR}) \quad (A.17)$$

and

$$C_P\dot{T}_P = q_{f,P} + \frac{1}{R_{A,V}}(T_A - T_P) - \frac{1}{R_{IR,P}}(T_P - T_{IR}) \quad (A.18)$$

. Rearranging the equations and choosing $x = [T_{IR}, T_C, T_V, T_P]^T$, $u = [q_{f,IR}, q_{f,C}, q_{f,V}, q_{f,P}]^T$ and $w = T_A$ result in the state-space representation given by

$$A = \begin{pmatrix} \frac{-1}{C_{IR}}\left(\frac{1}{R_{IR,C}} + \frac{1}{R_{IR,V}} + \frac{1}{R_{IR,P}}\right) & \frac{1}{C_{IR}R_{IR,C}} & \frac{1}{C_{IR}R_{IR,V}} & \frac{1}{C_{IR}R_{IR,P}} \\ \frac{1}{C_C R_{IR,C}} & \frac{-1}{C_C}\left(\frac{1}{R_{A,C}} + \frac{1}{R_{IR,C}}\right) & 0 & 0 \\ \frac{1}{C_V R_{IR,V}} & 0 & \frac{-1}{C_V}\left(\frac{1}{R_{A,V}} + \frac{1}{R_{IR,V}}\right) & 0 \\ \frac{1}{C_P R_{IR,P}} & 0 & 0 & \frac{-1}{C_P}\left(\frac{1}{R_{A,P}} + \frac{1}{R_{IR,P}}\right) \end{pmatrix} \quad (A.19)$$

$$B = \begin{pmatrix} \frac{1}{C_{IR}} & 0 & 0 & 0 \\ 0 & \frac{1}{C_C} & 0 & 0 \\ 0 & 0 & \frac{1}{C_V} & 0 \\ 0 & 0 & 0 & \frac{1}{C_P} \end{pmatrix} \quad (A.20)$$

$$N = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{1}{C_C R_{A,C}} & 0 & 0 \\ 0 & 0 & \frac{1}{C_V R_{A,V}} & 0 \\ 0 & 0 & 0 & \frac{1}{C_P R_{A,P}} \end{pmatrix} \quad (A.21)$$

and

$$C = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (A.22)$$

Bibliography

- [1] In Milan Brandt, editor, *Laser Additive Manufacturing*, Woodhead Publishing Series in Electronic and Optical Materials, page ii. Woodhead Publishing, 2017. ISBN 978-0-08-100433-3. doi: <https://doi.org/10.1016/B978-0-08-100433-3.09001-1>. URL <http://www.sciencedirect.com/science/article/pii/B9780081004333090011>.
- [2] Isizoh A N, Okide S.O, Anazia A.E, and Ogu C.D. Temperature control system using fuzzy logic technique. *International Journal of Advanced Research in Artificial Intelligence*, 1:27–30, 06 2012. doi: 10.14569/IJARAI.2012.010305.
- [3] H. Bikas, P. Stavropoulos, and G. Chrysosolouris. Additive manufacturing methods and modelling approaches: a critical review. *The International Journal of Advanced Manufacturing Technology*, 83(1):389–405, Mar 2016. ISSN 1433-3015. doi: 10.1007/s00170-015-7576-2. URL <https://doi.org/10.1007/s00170-015-7576-2>.
- [4] David L. Bourell, Trevor J. Watt, David K. Leigh, and Ben Fulcher. Performance limitations in polymer laser sintering. *Physics Procedia*, 56:147 – 156, 2014. ISSN 1875-3892. doi: <https://doi.org/10.1016/j.phpro.2014.08.157>. URL <http://www.sciencedirect.com/science/article/pii/S1875389214003022>. 8th International Conference on Laser Assisted Net Shape Engineering LANE 2014.
- [5] M. Brandt. The role of lasers in additive manufacturing. In Milan Brandt, editor, *Laser Additive Manufacturing*, Woodhead Publishing Series in Electronic and Optical Materials, pages 1 – 18. Woodhead Publishing, 2017. ISBN 978-0-08-100433-3. doi: <https://doi.org/10.1016/B978-0-08-100433-3.02001-7>. URL <http://www.sciencedirect.com/science/article/pii/B9780081004333020017>.
- [6] T. H. C. Childs, M. Berzins, G. R. Ryder, and A. Tontowi. Selective laser sintering of an amorphous polymer—simulations and experiments. *Proceedings of the Institution of Mechanical Engineers, Part B: Journal of Engineering*

- ing Manufacture*, 213(4):333–349, 1999. doi: 10.1243/0954405991516822. URL <https://doi.org/10.1243/0954405991516822>.
- [7] H. Czichos, T. Saito, and L.E. Smith. *Springer Handbook of Materials Measurement Methods*. Springer Berlin Heidelberg, 2007. ISBN 9783540303008. URL <https://books.google.se/books?id=81ANaR-Pqi4C>.
- [8] M. Dressler, M. Röllig, M. Schmidt, A. Maturilli, and J. Helbert. Temperature distribution in powder beds during 3D printing. *Rapid Prototyping Journal*, 16:328–336, 08 2010. doi: 10.1108/13552541011065722.
- [9] Martin Enqvist, Torkel Glad, Svante Gunnarsson, Peter Lindskog, Lennart Ljung, Johan Löfberg, Tomas McKelvey, Anders Stenman, and Jan-Erik Strömberg. *Industriell reglerteknik - Kurskompendium*. Reglerteknik, Institutionen för systemteknik, Linköpings universitet, Linköping, Sweden, 2014.
- [10] Torkel Glad and Lennart Ljung. *Reglerteknik - Grundläggande teori*. Studentlitteratur AB, Lund, Sweden, 2016.
- [11] Torkel Glad and Lennart Ljung. *Reglerteori - Flervariabla och olinjära metoder*. Studentlitteratur AB, Lund, Sweden, 2016.
- [12] R. Goodridge and S. Ziegelmeier. Powder bed fusion of polymers. In Milan Brandt, editor, *Laser Additive Manufacturing*, Woodhead Publishing Series in Electronic and Optical Materials, pages 181 – 204. Woodhead Publishing, 2017. ISBN 978-0-08-100433-3. doi: <https://doi.org/10.1016/B978-0-08-100433-3.00007-5>. URL <http://www.sciencedirect.com/science/article/pii/B9780081004333000075>.
- [13] Fredrik Gustafsson. *Statistical Sensor Fusion*. Studentlitteratur AB, Lund, Sweden, 2018.
- [14] Fredrik Gustafsson, Lennart Ljung, and Mille Millnert. *Signal Processing*. Studentlitteratur AB, Lund, Sweden, 2011.
- [15] Lennart Ljung. *System Identification - Theory for the user - Second edition*. Prentice-Hall, Inc., New Jersey, USA, 2012.
- [16] Lennart Ljung and Torkel Glad. *Modellbygge och simulering*. Studentlitteratur AB, Lund, Sweden, 2011.
- [17] Matlab. Matlab system identification toolbox, 2019. URL <https://se.mathworks.com/products/sysid.html>.
- [18] University of Roma. Thermal-electrical analogy: thermal network, 2019. URL http://www.ingaero.uniroma1.it/attachments/2176_Cap_3%20Thermal-electrical%20analogy.pdf.
- [19] Mikael Olofsson. *Signal Theory*. Studentlitteratur AB, Lund, Sweden, 2011.

- [20] E. Ramirez-Laboreo, C. Sagues, and S. Llorente. Thermal modeling, analysis and control using an electrical analogy. pages 505–510, 11 2014. doi: 10.1109/MED.2014.6961423.
- [21] Edgar Ramirez-Laboreo, Carlos Sagues, and Sergio Llorente. Dynamic heat and mass transfer model of an electric oven for energy analysis. *Applied Thermal Engineering*, 93:683 – 691, 2016. ISSN 1359-4311. doi: <https://doi.org/10.1016/j.applthermaleng.2015.10.040>. URL <http://www.sciencedirect.com/science/article/pii/S135943111501087X>.
- [22] D. Riedlbauer, M. Drexler, D. Drummer, P. Steinmann, and J. Mergheim. Modelling, simulation and experimental validation of heat transfer in selective laser melting of the polymeric material PA12. *Computational Materials Science*, 93:239 – 248, 2014. ISSN 0927-0256. doi: <https://doi.org/10.1016/j.commatsci.2014.06.046>. URL <http://www.sciencedirect.com/science/article/pii/S0927025614004571>.
- [23] Manfred Schmid. Ls materials: Polymer powders. In Manfred Schmid, editor, *Laser Sintering with Plastics*, pages 101 – 124. Hanser, 2018. ISBN 978-1-56990-683-5. doi: <https://doi.org/10.3139/9781569906842.005>. URL <http://www.sciencedirect.com/science/article/pii/B9781569906835500065>.
- [24] Manfred Schmid. Ls materials: Commercial materials. In Manfred Schmid, editor, *Laser Sintering with Plastics*, pages 125 – 159. Hanser, 2018. ISBN 978-1-56990-683-5. doi: <https://doi.org/10.3139/9781569906842.006>. URL <http://www.sciencedirect.com/science/article/pii/B9781569906835500077>.
- [25] Manfred Schmid. Ls materials: Polymer properties. In Manfred Schmid, editor, *Laser Sintering with Plastics*, pages 65 – 99. Hanser, 2018. ISBN 978-1-56990-683-5. doi: <https://doi.org/10.3139/9781569906842.004>. URL <http://www.sciencedirect.com/science/article/pii/B9781569906835500053>.
- [26] B. Sohlberg. Grey box modelling for model predictive control of a heating process. *Journal of Process Control*, 13(3):225 – 238, 2003. ISSN 0959-1524. doi: [https://doi.org/10.1016/S0959-1524\(02\)00030-6](https://doi.org/10.1016/S0959-1524(02)00030-6). URL <http://www.sciencedirect.com/science/article/pii/S0959152402000306>.
- [27] S. Sun, M. Brandt, and M. Easton. Powder bed fusion processes: An overview. In Milan Brandt, editor, *Laser Additive Manufacturing*, Woodhead Publishing Series in Electronic and Optical Materials, pages 55 – 77. Woodhead Publishing, 2017. ISBN 978-0-08-100433-3. doi: <https://doi.org/10.1016/B978-0-08-100433-3.00002-6>. URL <http://www.sciencedirect.com/science/article/pii/B9780081004333000026>.